Sleep, Physical Activity, and Acid-Related Upper-GI Disease

Gastroesophageal Reflux Disease and Oesophagitis in the *All of Us* Research Program (Controlled Tier v9)

A Technical Report on Study Design, Data Engineering, and Statistical Modeling

Functional GI Disorders & Physiological Measures Project

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Abstract

This report documents the design and implementation of a retrospective, cross-sectional analysis relating Fitbit-recorded *sleep* and *physical-activity* metrics (exposures) to two electronic-health-record (EHR) outcomes — gastroesophageal reflux disease (`has_gerd_any`, seed concept 318800) and *oesophagitis* (`has_esophagitis`, seed concept 30753) — among participants of the *All of Us* Research Program (Controlled Tier CDR v9, C2024Q3R9) who shared both wearable and EHR data. The study is organized as a 2×3 matrix: two outcomes crossed with three exposure sets (sleep, activity, and a combined activity+sleep set), giving six analysis configurations executed by one shared statistical engine. Six new R scripts implement the work without modifying any existing notebook: a dual-phenotype GERD outcome pull, a shared modeling-engine helper, three analysis files (sleep, activity, combined), and an optional proton-pump-inhibitor / H_2-antagonist pull supporting a medication-treated "severe GERD" sensitivity analysis. This document details the phenotype definitions, cohort-eligibility logic for both wearable streams, exposure quartilization, covariate engineering (including the GERD-specific peptic-ulcer-disease and IBS covariates), the centralized descriptive / univariable / multivariable logistic-regression engine, mutually-adjusted combined models, assumption diagnostics, sensitivity analyses, and the ways the GERD pipeline both re-uses and departs from the project's IBS and constipation templates. It is intended as a self-contained methodological reference for reproducing and critiquing the study.

Deliverables. `GERD Outcome Data Upload R9.R`, `GERD Analysis Helpers R9.R`, `GERD Concepts R9.R`, `GERD Build Cohort From Export.R`, `GERD Analysis Helpers R9.R`, `GERD Data Prep R9.R`, `RUN_GERD_ANALYSIS.R`, `Sleep and GERD Analysis File R9.R`, `Activity and GERD Analysis File R9.R`, `Activity Sleep and GERD Analysis File R9.R`, `GERD Spline Analysis R9.R`, `GERD Quartile vs Spline Comparison R9.R`, `RUN_GERD_SPLINE_ANALYSIS.R`, `GERD Pharmacologics Data Upload.R`.

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1 Executive Summary

This report specifies a retrospective, cross-sectional analysis conducted in the All of Us Research Program (Controlled Tier Curated Data Repository v9, release C2024Q3R9) that relates objectively measured, Fitbit-recorded physiological exposures – sleep and physical activity – to acid-related upper gastrointestinal disease ascertained from electronic health records (EHR). The study is deliberately built around *two* binary outcome phenotypes and *three* exposure sets, yielding a 2×3 grid of six primary analysis configurations that share a single statistical engine.

The two outcomes are EHR phenotypes, each requiring at least two condition records (≥ 2) and each drawn from its own dedicated, descendant-expanded condition pull. `has_gerd_any` captures gastroesophageal reflux disease (seed concept **318800**), the broad group dominated by symptom-based diagnosis. `has_esophagitis` captures oesophagitis (seed concept **30753**, with erosive oesophagitis **4231067** folded in), the phenotype carrying documented mucosal injury. The two are not mutually exclusive — a participant may accumulate codes from both over time, and the pipeline quantifies that overlap explicitly rather than making it a third outcome. Analysing them side by side is what allows the study to ask whether physiological exposures act on symptom perception, on measurable tissue damage, or on both.

The three exposure sets are each analyzed against *both* outcomes. (A) Sleep exposures are cohort quartiles (with Q1 as the reference level) of average nightly minutes asleep, minutes in bed, minutes awake, minutes restless, deep, light, and REM sleep, and sleep efficiency (asleep divided by in-bed). (B) Activity exposures are cohort quartiles of average daily steps, lightly, fairly, very, and total active minutes, sedentary minutes, and a corrected maximum daily heart rate. (C) The combined set operates on the intersection cohort (participants with valid sleep *and* valid activity streams) and analyzes each of the roughly fifteen exposures separately, plus mutually-adjusted models that carry a sleep quartile and an activity quartile together in the same equation (for example, minutes-asleep quartile jointly with steps quartile).

Exposure set	Outcome `has_gerd_any`	Outcome `has_esophagitis`
(A) Sleep quartiles	Config 1	Config 2
(B) Activity quartiles	Config 3	Config 4
(C) Combined (+ mutually adjusted)	Config 5	Config 6

Table 1: The six analysis configurations formed by crossing three exposure sets with the two parallel outcome phenotypes. Every configuration is fit with the identical modeling engine.

To guarantee that all six configurations use identical statistics, definitions, and reporting, the design centralizes methodology in a shared engine file, "GERD Analysis Helpers R9.R". That helper defines the outcome objects and factor labels (`GERD_OUTCOMES` with "No GERD"/"GERD" and "No oesophagitis"/"Oesophagitis"), the exposure lists (`SLEEP_EXPOSURES`, `ACTIVITY_EXPOSURES`), the adjustment set (`GERD_ADJ_COVARS`, which adds the reflux-specific and expert-required terms described in the Covariates section), the master label lookup (`GERD_LABELS`), and the reusable functions `prep_modeling_df()`, `descriptive_table()`, `univariate_table()`, `run_models_for_outcome()`, and `run_diagnostics()`. A single dispatcher, `analyze_all_outcomes()`, loops over both outcomes so that each analysis file merely builds its data frame, sources the helper, and calls the dispatcher.

The deliverables are all new R files; no existing notebook is edited:

- "GERD Outcome Data Upload R9.R" – two separate descendant-expanded outcome pulls, one per phenotype (seed 318800 and seed 30753), producing `R9_gerd_all_outcome.rds` and `R9_esophagitis_outcome.rds`, plus an overlap report.
- "GERD Analysis Helpers R9.R" – the shared modeling engine described above.

- “Sleep and GERD Analysis File R9.R” – sleep exposures against both outcomes.
- “Activity and GERD Analysis File R9.R” – activity exposures against both outcomes.
- “Activity Sleep and GERD Analysis File R9.R” – the combined cohort, both outcomes, plus mutually-adjusted models.
- “GERD Pharmacologics Data Upload.R” – an optional PPI/H2RA prescription pull supporting a medication-treated (“severe”) GERD sensitivity analysis, producing `R9_gerd_drug_df.rds`.

Primary analytic outputs are `R9_final_analysis_sleep_gerd_df.rds`, `R9_final_analysis_activity_gerd_df.rds`, and `R9_final_analysis_activity_sleep_gerd_df.rds`, together with the reusable status objects `R9_gerd_status.rds` and `R9_esophagitis_status.rds`. All estimates are adjusted odds ratios from logistic regression, quartile exposures are compared against Q1, and multiplicity across the many exposure–outcome contrasts is controlled with the Bonferroni correction (matching the antecedent IBS paper; configurable to FDR). Because the design is cross-sectional, the primary estimates are associations; a pre-specified post-Fitbit temporal outcome definition serves as the strongest guard against reverse causation.

2 Introduction and Scientific Background

2.1 Acid-Related and Functional Upper Gastrointestinal Disorders

Gastroesophageal reflux disease (GERD) is among the most prevalent chronic gastrointestinal conditions in adults, defined clinically as troublesome symptoms or mucosal complications arising when gastric contents reflux into the oesophagus. It sits at the intersection of two overlapping disease constructs. On one side are the acid-related structural disorders, in which excess oesophageal acid exposure produces objective epithelial injury – erosions, ulceration, stricture, and Barrett metaplasia. On the other side are the functional and symptom-defined presentations, in which patients experience heartburn, regurgitation, and non-cardiac chest pain in the absence of, or out of proportion to, visible mucosal damage. This latter category shares mechanistic and epidemiologic features with the functional gastrointestinal disorders (now termed disorders of gut–brain interaction), including a prominent role for visceral hypersensitivity, altered central pain processing, and psychosocial and behavioral modulation of symptom burden.

This dual character motivates the study’s two parallel phenotypes, which are deliberately drawn on either side of the erosive divide. Broad GERD (`has_gerd_any`) is diagnosed predominantly on symptoms in the absence of documented mucosal damage, and therefore behaves largely as a disorder of gut–brain interaction. Oesophagitis (`has_esophagitis`) is defined by objective tissue injury. Studying both in parallel lets us ask whether physiological exposures act on subjective symptom reporting, on measurable tissue damage, or on both, and whether the strength of association differs between the sensory/perceptual and the structural manifestations of reflux.

2.2 The Gut-Brain Axis and Behavioral-Physiological Exposures

The gut–brain axis is the bidirectional signaling network linking the central nervous system, the autonomic nervous system, the enteric nervous system, and the gut. Through this axis, higher-order behavioral states – sleep, physical activity, arousal, and stress – exert continuous influence over gastrointestinal motility, secretion, mucosal blood flow, and afferent pain signaling, while visceral signals in turn shape sleep architecture and perceived exertion. Sleep and physical activity are therefore not incidental lifestyle covariates but plausible upstream physiological modulators of reflux pathophysiology.

Historically, these exposures have been measured by self-report, which is subject to recall bias, social desirability, and poor correspondence with objectively recorded behavior. Consumer wearable devices such as Fitbit allow longitudinal, objective, per-night and per-day quantification of sleep quantity and quality and of ambulatory activity intensity and volume. This study exploits that measurement richness: rather than a single crude exposure, it decomposes sleep into duration, continuity, and stage architecture (deep, light, REM, awake, restless, efficiency) and activity into volume (steps), intensity strata (lightly, fairly, very active minutes), sedentary time, and a fitness/exertion proxy (corrected maximum daily heart rate). Quartile categorization (Q1 reference) preserves monotone dose–response signal while remaining robust to the skew and device-specific scaling of these metrics.

2.3 Mechanisms Linking Poor Sleep and Low Physical Activity to Reflux Disease

Several partially overlapping mechanisms make both insufficient or fragmented sleep and low physical activity biologically plausible contributors to GERD and, more specifically, to erosive oesophagitis.

Supine nocturnal acid exposure. Sleep is spent predominantly recumbent, and the supine posture removes the gravitational assistance that clears refluxate during waking hours. Nocturnal reflux episodes tend to be longer and to reach more proximal segments of the oesophagus, prolonging contact time between acid and the mucosa. Individuals with shorter, more fragmented, or more inefficient sleep may accumulate greater cumulative supine acid-contact time, and disturbed sleep architecture is itself associated with a higher frequency of nocturnal reflux events. Because prolonged acid contact is the proximate driver of epithelial erosion, this pathway predicts a particularly strong link to the erosive `has_esophagitis` phenotype.

Delayed acid clearance. Oesophageal acid clearance depends on primary peristalsis triggered by swallowing and on neutralization by swallowed, bicarbonate-rich saliva. Both swallowing frequency and salivary flow fall markedly during sleep, and this suppression is exaggerated by sleep deprivation and fragmentation. The result is slower restoration of normal intraluminal pH after a reflux event, again lengthening mucosal acid exposure. Reduced daytime physical activity may compound impaired motility indirectly through its downstream metabolic and autonomic effects.

Visceral hypersensitivity. Experimental sleep deprivation lowers the oesophageal pain threshold, producing hyperalgesia to acid and mechanical distension. This heightened afferent sensitivity increases symptom reporting for a given quantum of acid exposure and is a core feature of the gut–brain-interaction component of reflux. This pathway predicts that poor sleep will associate with symptom-defined `has_gerd_any` even where objective injury is modest, and it is a mechanism through which sleep can influence perceived disease independently of tissue damage.

Autonomic and vagal tone. Both habitual physical activity and healthy sleep shape autonomic balance, and vagal tone modulates lower oesophageal sphincter (LES) pressure and the frequency of transient LES relaxations (TLESRs), the dominant mechanism of reflux events. Disrupted sleep and physical deconditioning shift autonomic balance in directions associated with impaired sphincter function and altered gastric accommodation and emptying. The corrected maximum daily heart rate is included in part as a proxy that carries autonomic and cardiorespiratory-fitness information alongside the volume and intensity activity metrics.

Adiposity and mechanical load. Low physical activity is a determinant of positive energy balance, weight gain, and especially central (visceral) adiposity. Increased intra-abdominal pressure raises the gastro-oesophageal pressure gradient, promotes hiatal hernia formation, and disrupts the anti-reflux barrier – a well-established mechanical route from sedentary behavior to reflux and to erosive disease. Adiposity is therefore both a mediator on the causal path from inactivity to GERD and a confounder to be handled analytically; the design measures median BMI (from measurement concept 3038553) and adjusts for it, recognizing that this partially blocks a genuine causal pathway and may attenuate activity–outcome estimates.

Reverse causation. Critically, the sleep–reflux relationship is bidirectional: nocturnal acid reflux itself provokes arousals and fragments sleep, so poor sleep can be a consequence rather than a cause of GERD. Likewise, symptomatic or severe reflux can discourage vigorous exertion, and post-prandial vigorous activity can transiently provoke reflux, so the activity–reflux relationship is not unidirectional either. In a cross-sectional design these bidirectional effects cannot be fully disentangled, which is why the analysis pre-specifies a post-Fitbit temporal outcome definition (first GERD code recorded ≥ 180 days after the first Fitbit date, then ≥ 2 codes) as its principal guard against reverse causation, complemented by distinct-date, overlap-set, and medication-treated sensitivity analyses.

2.4 Why Erosive Oesophagitis Is a Meaningful Stricter Phenotype

Broad GERD ascertained from EHR condition codes is heterogeneous: it aggregates objective erosive disease with a large fraction of symptom-driven, non-erosive, and empirically treated reflux, and it is susceptible to differential coding by healthcare utilization and to misclassification of functional heartburn as GERD. Erosive oesophagitis (`has_esophagitis`), by contrast, requires documentation consistent with visible mucosal injury and thus anchors the analysis to a more objective, endoscopically grounded endpoint. Several considerations make it a valuable second phenotype rather than a redundant one.

First, the two phenotypes plausibly load on different mechanisms: symptom-defined GERD is more strongly shaped by visceral hypersensitivity and central pain processing (sensitive to sleep-related hyperalgesia), whereas erosive disease is more directly a function of cumulative acid-contact time and barrier failure (sensitive to supine exposure, delayed clearance, and adiposity). Contrasting the exposure associations across the two outcomes is therefore mechanistically informative. Second, erosive oesophagitis is less vulnerable to the diagnostic-labeling noise that inflates symptom-coded GERD, so a concordant signal across both phenotypes strengthens causal interpretation, while a signal confined to symptom-defined GERD points toward a perceptual rather than a structural pathway. Third, because the two phenotypes are drawn from separate pulls rather than nested, a contrast between them is a contrast between distinct clinical entities rather than between a set and its own subset — which makes any divergence in exposure association directly interpretable. Oesophagitis is nonetheless expected to be the rarer of the two, which reduces power and widens confidence intervals; the analysis accommodates this through complete-case reporting, multiplicity control, and explicit diagnostic checks for separation and unstable standard errors. Reporting both phenotypes with the identical modeling engine makes the comparison fair and transparent.

2.5 Rationale and the All of Us Research Opportunity

Prior evidence linking sleep and activity to reflux is dominated by cross-sectional self-report, small clinical samples, and endpoints that rarely separate symptom-defined from erosive disease. The All of Us Research Program provides a distinctive opportunity to address these gaps at scale: it links longitudinal, device-recorded Fitbit sleep and activity data to a rich, standardized (OMOP-mapped) EHR and to detailed survey and physical-measurement data across a large, deliberately diverse participant population. This enables objectively measured exposures, EHR-based outcome phenotyping with an explicit $\geq$ 2-record rule, adjustment for an extensive covariate set (demographics, socioeconomic factors, alcohol, smoking, BMI, and a panel of comorbidities and medication-use flags), and reflux-specific confounders including peptic ulcer disease (`has_pud`), obstructive sleep apnoea (`has_sleep_apnea`) and sleep medication (`on_sleep_med`).

The design mirrors the project's antecedent IBS sleep study, constipation activity study, and constipation outcome upload, but adapts them for reflux: it swaps the IBS outcome for the GERD outcomes and adds the erosive subset, sources outcomes from a dedicated GERD pull (necessary because `R9_condition_df.rds` does not contain GERD), removes all IBS-subtype and multinomial logic, adds `has_pud`, `has_sleep_apnea`, `on_sleep_med` and `on_narcotic` as covariates, harmonizes `R9_*` file naming, extends the work to activity and combined analyses, and centralizes all statistics in the shared helper so that every configuration is estimated identically.

2.6 Research Question and Directional Hypothesis

The central research question is whether objectively measured sleep and physical activity are associated with the odds of broad GERD and of erosive oesophagitis in Fitbit-sharing All of Us participants, and whether any such associations persist after adjustment for demographic, socioeconomic, anthropometric, comorbidity, and medication confounders. The directional hypothesis is that greater sleep quantity and quality (for example, higher minutes asleep, higher sleep efficiency, and more consolidated sleep) and greater physical activity (for example, higher daily steps and more active minutes, with correspondingly lower sedentary time) associate with *lower* odds of both GERD and oesophagitis – that is, upper quartiles (Q2–Q4) will carry adjusted odds ratios below unity relative to Q1. We anticipate that these associations will attenuate after covariate adjustment – particularly for activity exposures, where BMI lies partly on the causal pathway – but that they will persist rather than disappear. Given the mechanistic profile described above, we expect the acid-exposure-dependent pathways to register on the erosive `has_esophagitis` phenotype, while sleep-related hyperalgesia may leave a signal on symptom-defined `has_gerd_any` even where mucosal injury is limited. Because the design is cross-sectional, all primary estimates are interpreted as associations; the post-Fitbit temporal outcome definition provides the strongest available guard against the reverse-causation pathway in which nocturnal reflux itself fragments sleep.

3 Study Design Overview

This study is a retrospective, cross-sectional analysis conducted in the All of Us Research Program using the Controlled Tier Curated Data Repository (CDR) v9, release C2024Q3R9. It relates wearable-recorded physiological exposures, derived from Fitbit devices, to acid-related upper gastrointestinal disease captured in the electronic health record (EHR). The analysis estimates associations between habitual sleep and physical-activity patterns and the odds of two parallel EHR-defined outcomes: gastroesophageal reflux disease (`has_gerd_any`) and the erosive subset (`has_esophagitis`). Because the design is cross-sectional, the primary estimands are associations rather than causal effects; a post-Fitbit temporal outcome definition is carried as the principal guard against reverse causation, since nocturnal reflux is itself known to fragment sleep.

The design is organized around a factorial matrix of two outcomes crossed with three exposure sets, yielding six analysis configurations that are executed by a common statistical engine. The two outcomes are binary EHR phenotypes requiring ≥ 2 qualifying condition records. The three exposure sets are (A) sleep metrics, (B) activity metrics, and (C) a combined activity-and-sleep set evaluated on the intersection cohort, including mutually adjusted models that carry one sleep and one activity exposure simultaneously. Every exposure is operationalized as a cohort quartile factor with the first quartile (Q1) as the reference level, and every configuration is fit with the identical multivariable logistic-regression specification described in Section 3.4 of the Methods (the shared modeling engine). The remainder of this section presents a design-at-a-glance summary, enumerates the six configurations and the deliverable code, and explains the don't-repeat-yourself (DRY) architecture that guarantees statistical identity across all six.

3.1 Design at a Glance

Table 2 summarizes the core design parameters. The study population is the set of Fitbit-sharing All of Us participants who are age ≥ 18 at their first relevant Fitbit date, who share EHR data (condition or measurement records), and who meet the stream-specific wear-time and valid-record thresholds detailed in the Data and Cohort sections. Exposures are cohort-internal quartiles so that each configuration compares the upper three quartiles of an exposure against its own Q1 reference within the analytic cohort for that exposure set.

Design element	Specification
Data source	All of Us Research Program, Controlled Tier CDR v9 (release C2024Q3R9)
Study design	Retrospective, cross-sectional, EHR outcomes with wearable-derived exposures
Population	Fitbit-sharers; age ≥ 18 at first Fitbit date; sharing EHR (condition or measurement)
Exposures	Cohort quartiles (Q1 reference) of Fitbit sleep and activity metrics; combined set adds mutually adjusted sleep + activity models
Outcomes	`has_gerd_any` (GERD, seed 318800) and `has_esophagitis` (oesophagitis, seed 30753); ≥ 2 records each, two separate pulls, parallel rather than nested
Primary model	Multivariable logistic regression; $\exp(\beta_j)$ is the adjusted odds ratio for quartile j versus Q1
Covariate anchor	Pre-Fitbit window; combined cohort anchored to the earliest Fitbit date across sleep and activity streams (pmin)
Covariates	Demographics, socioeconomic status, median BMI, comorbidities incl. `has_pud` and `has_sleep_apnea`, and medication-use flags incl. `on_sleep_med` and `on_narcotic`
Missing data	Binary EHR indicators coalesced NA $\rightarrow$ FALSE; survey/measurement missingness retained as an explicit Missing level
Multiplicity	Bonferroni correction (paper-matching; FDR selectable)
Configurations	Six: 2 outcomes $\times$ 3 exposure sets, all fit by one shared engine
Sensitivity analyses	Distinct-date GERD; post-Fitbit temporal GERD; sleep $\times$ activity overlap set; optional medication-treated (severe) GERD

Table 2: Design at a glance for the GERD and oesophagitis wearable-exposure analysis.

3.2 The Six Analysis Configurations

The analytic space is the 2×3 crossing of outcomes and exposure sets shown in Table 3. Rows correspond to the two parallel outcomes and columns correspond to the three exposure sets. Each of the three exposure sets is implemented by a single analysis file that builds one modeling data frame and then loops over both outcomes via the shared driver `analyze_all_outcomes()`; consequently the two configurations in a given column share the same analysis file and the same primary output data frame, differing only in which outcome column the engine stratifies and models. This is why the file and primary output are common within a column: the outcome loop is internal to a single script, not split across scripts.

Outcome		(A) Sleep exposures	(B) Activity exposures	(C) Combined (activity + sleep)
`has_gerd_any`	(GERD)	**C1**. File `Sleep and GERD Analysis File R9.R`; output `R9_final_analysis_sleep_gerd_df.rds`	**C3**. File `Activity and GERD Analysis File R9.R`; output `R9_final_analysis_activity_gerd_df.rds`	**C5**. File `Activity Sleep and GERD Analysis File R9.R`; output `R9_final_analysis_activity_sleep_gerd_df.rds`
`has_esophagitis`	(erosive subset)	**C2**. Same file and primary output as C1 (second outcome in the loop)	**C4**. Same file and primary output as C3 (second outcome in the loop)	**C6**. Same file and primary output as C5 (second outcome in the loop)

Table 3: The six analysis configurations (rows: outcomes; columns: exposure sets), naming the analysis file and primary output data frame for each. Both configurations in a column are produced by the same file because `analyze_all_outcomes()` loops both outcomes over one modeling data frame.

The combined column (C5, C6) is defined on the intersection cohort of participants who satisfy both the sleep-validity threshold (≥ 180 valid nights) and the activity-validity threshold (≥ 30 valid days). In that column each of the roughly fifteen exposures is analyzed on its own and, in addition, mutually adjusted models are fit that include one sleep-metric quartile and one activity-metric quartile together (for example, minutes-asleep quartile jointly with steps quartile). Covariate timing for the combined cohort is anchored to each participant's earliest Fitbit date across the two streams, computed as the element-wise minimum (pmin) of the first sleep date and the first activity date, so that all pre-Fitbit covariate windows are defined consistently regardless of which stream began first.

3.3 Deliverable Files and Their Roles

The project is delivered as six new R scripts, listed in Table 4. Two are data-upload scripts that materialize outcome and pharmacologic data via BigQuery / Cohort Builder pulls; one is the shared modeling engine; and three are the analysis files that build modeling data frames and invoke the engine. The primary modeling outputs are the three `R9_final_analysis_*_gerd_df.rds` data frames plus the reusable outcome-status objects `R9_gerd_status.rds` and `R9_esophagitis_status.rds`.

A dedicated outcome pull is required because the existing comorbidity object does not contain GERD condition records. Each outcome is seeded on its own concept and expanded to all standard descendants. Because each pull is already restricted to its own concept set, the phenotype is computed over all rows of that file rather than by re-filtering on the seed identifier; re-filtering on the seed alone would incorrectly drop descendant-coded cases. Membership therefore depends only on which file a record is in, rather than on a concept-name match within a single combined pull.

3.4 Shared-Helper Architecture and Statistical Identity

All statistical logic is centralized in a single helper file, `GERD Analysis Helpers R9.R`, which each analysis file loads with `source()` before calling the common driver. This don't-repeat-yourself (DRY) architecture exists specifically to guarantee that all six configurations are estimated with byte-for-byte identical statistics: the only quantities allowed to differ across configurations are the input modeling data frame and the exposure list handed to the engine.

The helper file is the single canonical definition of every modeling choice. It defines:

Deliverable file	Role
`GERD Outcome Data Upload R9.R`	Two separate descendant-expanded pulls, seeds 4144111 (GERD without oesophagitis) and 30753 (oesophagitis), each expands to all standard descendants, restricted to Fitbit-sharers, producing `R9_gerd_outcome.rds`, `R9_gerd_status.rds`, and `R9_esophagitis_status.rds`
`GERD Analysis Helpers R9.R`	Shared modeling engine (DRY): defines outcomes, exposure lists, adjustment covariates, label lookups, and all descriptive, univariate, model-fitting, and diagnostic functions
`Sleep and GERD Analysis File R9.R`	Builds the sleep modeling data frame and runs both outcomes; primary output `R9_final_analysis_sleep_gerd_df.rds` (configurations C1, C2)
`Activity and GERD Analysis File R9.R`	Builds the activity modeling data frame and runs both outcomes; primary output `R9_final_analysis_activity_gerd_df.rds` (configurations C3, C4)
`Activity Sleep and GERD Analysis File R9.R`	Builds the combined intersection-cohort data frame, runs both outcomes and mutually adjusted models; primary output `R9_final_analysis_activity_sleep_gerd_df.rds` (configurations C5, C6)
`GERD Pharmacologics Data Upload.R`	Optional PPI (ancestor 21600095) and H2RA (ancestor 21600096) prescription pull producing `R9_gerd_drug_df.rds`, used only by the medication-treated (severe) GERD sensitivity analysis and guarded by file existence

Table 4: Deliverable code files and their roles. All six files are new; no existing All of Us notebook is edited.

- `GERD_OUTCOMES` – the two outcomes together with their factor labels (“No GERD”/“GERD” and “No oesophagitis”/“Oesophagitis”), so outcome coding is fixed once.
- `SLEEP_EXPOSURES` and `ACTIVITY_EXPOSURES` – the canonical exposure inventories consumed by the sleep, activity, and combined files.
- `GERD_ADJ_COVARS` – the shared adjustment set (which adds `has_pud` and the expert-required terms relative to the sibling IBS study; the partial comorbidity Index by default), so every model adjusts for the same covariates.
- `GERD_LABELS` – the master label lookup used in all output tables.
- `prep_modeling_df()` – builds the collapsed factor covariates and relevels each quartile exposure to the Q1 $\rightarrow$ Q4 ordering with Q1 as reference.
- `descriptive_table()`, `univariate_table()`, `run_models_for_outcome()`, `run_diagnostics()`, and `analyze_all_outcomes()` – the descriptive, unadjusted, adjusted-model, diagnostic, and both-outcome driver routines.

Because these objects and functions live in exactly one place, every configuration inherits the same estimator (logistic regression), the same covariate specification, the same reference-level convention, the same complete-case handling, the same full-model-plus-backward-AIC pairing (via `MASS::stepAIC` with the exposure forced to remain), the same area-under-the-curve computation via `pROC`, the same variance-inflation-factor computation via `car`, the same descriptive stratified tables with chi-square tests and adjusted p-values, and the same diagnostic battery (independence, multicollinearity, separation and large standard errors, Cook's distance, and Box–Tidwell linearity for BMI). The driver `analyze_all_outcomes()` loops both outcomes so that a single call in each analysis file produces both configurations in that exposure column under identical code paths.

This centralization delivers three concrete guarantees. First, statistical identity: any two configurations that should be comparable are, by construction, computed by the same functions with the same defaults, eliminating silent divergence between the sleep, activity, and combined analyses. Second, single-point maintainability: a change to the covariate set, a label, or a diagnostic propagates to all six configurations at once, so the report cannot fall out of internal consistency. Third, auditability: the exact model that produced any table can be traced to one helper definition rather than to per-file copies that might drift. The design mirrors the project's IBS sleep study, constipation activity study, and constipation outcome upload, but removes all IBS-subtype logic (subtype flags and multinomial models), swaps the IBS covariate for the GERD outcomes plus the added `has_esophagitis`, adds the reflux-specific covariates, standardizes `R9_` naming, extends the analysis to activity and combined exposures, and centralizes the modeling engine in the shared helper.

Finally, all deliverables are new files. No existing All of Us notebook or script is modified: the outcome and pharmacologic pulls, the shared helper, and the three analysis files are added alongside the existing project code, and prior objects such as `R9_condition_df.rds` are read but never rewritten. This isolation keeps the GERD analysis reproducible and non-destructive with respect to the established IBS and constipation pipelines from which it inherits its templates.

4 Biological Mechanisms Linking Sleep and Activity to Reflux Disease

Because the study relates two behavioral-physiological exposures to two reflux phenotypes, it is worth setting out, in one place, the candidate mechanisms that make each exposure–outcome link plausible. These mechanisms also motivate the covariate set (each major confounder sits on one of these pathways) and the directional predictions of Section 19.

4.1 Lower-esophageal-sphincter tone and transient relaxations

Reflux events are driven largely by transient lower-esophageal-sphincter relaxations. Sleep deprivation, sympathetic arousal, and the substances associated with poor sleep and sedentary lifestyles (nicotine, alcohol, late large meals) each lower resting sphincter tone or increase the frequency of transient relaxations, raising the probability that gastric contents cross into the oesophagus. Adjusting for smoking and alcohol partially isolates the sleep/activity signal from these co-travelling behaviors.

4.2 Acid clearance and supine time

Once refluxate reaches the oesophagus, clearance depends on primary peristalsis (swallowing) and salivary bicarbonate, both of which are suppressed during sleep. Fragmented or short sleep can extend supine, low-clearance time and prolong mucosal acid contact, a pathway that plausibly matters more for the erosive oesophagitis phenotype (`has_esophagitis`) than for symptom-coded GERD, because sustained contact time is what produces mucosal injury.

4.3 Intra-abdominal pressure and adiposity

Higher body-mass index raises intra-abdominal pressure, promotes hiatal herniation, and increases the gastro-oesophageal pressure gradient favoring reflux. Adiposity is also associated with shorter, more fragmented sleep and with lower physical activity, making it a central confounder — which is why median BMI is included in every adjusted model and why any residual sleep/activity association is interpreted as over-and-above the adiposity pathway.

4.4 Autonomic and vagal regulation

Sleep and exercise both modulate autonomic balance. Vagal tone governs gastric motility and lower-esophageal-sphincter function; poor sleep shifts autonomic balance sympathetically, and habitual activity raises vagal tone. The corrected maximum-heart-rate exposure in the activity set is, in part, a proxy for this autonomic/fitness axis.

4.5 Visceral hypersensitivity and central pain processing

Sleep deprivation lowers visceral pain thresholds, so the same acid exposure can produce more symptoms. This pathway acts preferentially on the symptom-defined broad GERD phenotype rather than on objective mucosal injury, and it is the principal reason the two outcomes are analyzed in parallel: an exposure that acts through perception should show a stronger association with `has_gerd_any` than with `has_esophagitis`, whereas one acting through acid contact time should show the reverse or an equal association.

4.6 Circadian, inflammatory, and reverse-causal loops

Circadian disruption alters gastric acid secretion and melatonin (which has mucosal-protective effects), and low-grade systemic inflammation associated with poor sleep and inactivity may sensitize the oesophageal mucosa. Finally, the loop is bidirectional: nocturnal reflux itself awakens patients and fragments sleep, which is precisely why the post-Fitbit temporal outcome is carried as the primary reverse-causation guard.

5 Prior Evidence and the Knowledge Gap

Associations between lifestyle factors and reflux have been studied for decades, but the evidence base has three recurring limitations that this study is designed to address. First, most prior work measures *sleep and activity by self-report*, which is subject to recall and social-desirability bias and correlates only modestly with objective wearable measurement; here both exposures are device-recorded at nightly and daily resolution over months. Second, reflux outcomes are often ascertained by *symptom questionnaire in modest single-center samples*, limiting both objectivity and power; here the outcome is EHR-defined across a large, national, demographically broad cohort, and the erosive oesophagitis subset provides an objective, mucosal-injury endpoint alongside the symptom-coded broad phenotype. Third, few datasets *link high-resolution wearable streams to longitudinal EHR* at scale; the *All of Us* Research Program is purpose-built for exactly this linkage.

The specific gap this study fills is therefore the near-absence of large-scale, objectively-measured sleep-and-activity analyses of GERD — and, in particular, the lack of any analysis that separates the symptom-coded and erosive phenotypes while adjusting for an unusually complete covariate set including adiposity, psychiatric comorbidity, peptic ulcer disease, and IBS. By running the identical engine over sleep, activity, and combined exposures and over both outcomes, the design also permits an internal triangulation — across exposure domains and across outcome phenotypes — that single-exposure, single-outcome studies cannot provide. No numeric findings are asserted here; the contribution of this report is the specification of that analysis.

6 Data Sources

6.1 The *All of Us* Controlled Tier, CDR v9

All queries and derived data use the Controlled Tier Curated Data Repository, version 9 (`C2024Q3R9`). The outcome and drug pulls pin the workspace CDR explicitly with `Sys.setenv(WORKSPACE_CDR = "fc-aou-cdr-prod-ct.C2024Q3R9")` so that concept expansions resolve against the intended release. All person-level data remain inside the secure *All of Us* workspace; only code is version-controlled, and this report contains no individual-level data.

6.2 The three Workbench datasets

The GERD analysis is built entirely from three datasets the study team exported to Workspace Resources. It reads nothing from the project's pre-existing `.rds` objects, which are the same objects behind the published IBS paper:

Dataset	Supplies
`Demographic_and_Fitbit_Data`	Demographics; Fitbit sleep, steps and heart rate (the exposures)
`EHR_Data`	Conditions, procedures and drug exposures (the outcomes, comorbidities and medications)
`Survery_Data`	Survey covariates and BMI
`Antidepressants`, `Antipsychotics`, `Beta_Blockers`, `Calcium_Blockers`, `Narcotics`	Medication covariates, one export per class

The five medication exports are *authoritative* for their class: membership of the `Narcotics` export establishes opioid exposure without depending on an ingredient name matching a pattern. Tricyclics are separated out of the antidepressant export by name, because the sleep-medication covariate requires that class specifically. Folders are located by name anywhere beneath `~/workspace`, since they have already been reorganised once during the study.
All three carry `person_id`, and the build merges on it. The crucial point is that **the GERD cohort is not the IBS cohort**: nothing is read from `rds_backup`, so the two studies share no denominator and the GERD numbers cannot inherit an IBS-era eligibility decision.

Two properties of the export that the build has to handle. First, the Workbench applies a `T_DISP_` convention: a column `X` holds a coded value and `T_DISP_X` the readable text. Reading `X` where a name is expected silently yields concept ids. Second, the pre-aggregated Fitbit tables (`sleepDailySummary`, `activitySummary`) are entirely NULL in this export, while the underlying `sleepLevel` table is fully populated. The nightly sleep metrics are therefore reconstructed from `sleepLevel`:

$$\text{minute\_asleep} = \text{asleep} + \text{deep} + \text{light} + \text{rem}, \qquad \text{minute\_awake} = \text{awake} + \text{wake},$$

with `minute_in_bed` the sum of every level. The first form covers stages-capable devices, the second the classic logs. Wear time, which the export has no column for, is recovered as the sum of `minute_in_zone` across heart-rate zones, so the $\geq$ 10-hour valid-day rule still applies. Heart-rate zone minutes stand in for the empty active-minute columns; they are *heart-rate* based rather than accelerometer based and are labelled as such.

6.3 Obtaining the outcome records

The body describes `GERD Outcome Data Upload R9.R`, which queries `fc-aou-cdr-prod-ct.C2024Q3R9` and exports through Cloud Storage. In the migrated workspace that project lies *outside* the VPC Service Controls perimeter, and every request fails with `policyViolation`. The replacement, `GERD_PULL_OUTCOMES_BIGQUERY.R`, resolves the BigQuery dataset attached to the workspace as a Resource (e.g. `C2025Q4R6`, which is inside the perimeter), expands descendants through `concept_ancestor` with a `cb_criteria` fallback, restricts to the Fitbit cohort's `person_ids`, and downloads directly — removing the Cloud Storage hop entirely. The original script is retained for workspaces with classic All of Us BigQuery access.

6.4 Wearable and EHR inputs

The analysis draws on three Fitbit streams and four EHR/survey domains. The sleep stream provides nightly stage minutes; the intraday-steps stream provides per-day step totals and wear hours and defines valid activity days; the daily activity stream provides activity-zone minutes; and a pre-computed corrected maximum-heart-rate summary supplies the cardiovascular exposure. The condition, measurement, drug, survey, and person tables supply the outcome (indirectly), covariates, and demographics. Table 5 lists every input.

Input `.rds`	**Contents / role**
`R9_gerd_outcome.rds`	**New.** All GERD condition records for Fitbit-sharers; source of both outcomes.
`R9_fitbit_sleep_daily_summary_df.rds`	Nightly Fitbit sleep summaries (sleep exposure source).
`R9_fitbit_activity_df.rds`	Daily activity-zone minutes (sedentary, lightly/fairly/very active).
`R9_fitbit_intraday_steps_df.rds`	Intraday steps and wear hours; defines valid activity days.
`R9_max_hr_minute_all_days_df.rds`	Corrected maximum daily heart rate (optional exposure).
`R9_person_df.rds`	Demographics (date of birth, sex at birth, race, ethnicity, gender).
`R9_survey_df.rds`	Survey answers (smoking, alcohol, education, income).
`R9_measurement_df.rds`	Measurements, including body mass index (concept 3038553).
`R9_drug_df.rds`	Drug exposures for cardiac/psychiatric medication covariates.
`R9_condition_df.rds`	Comorbidity conditions (and IBS); *does not* contain GERD.
`R9_pud_status.rds`	Pre-computed peptic-ulcer-disease covariate (`has_pud`).
`cci_scored_sleep.rds` / `R9_cci_scored.rds`	Charlson Comorbidity Index scores.
`R9_gerd_drug_df.rds`	**New, optional.** PPI/H_2RA exposures (treated-GERD analysis).

Table 5: Datasets consumed by the GERD analyses. Only `R9_gerd_outcome.rds` (and optionally `R9_gerd_drug_df.rds`) must be created; everything else is reused as-is.

6.5 A key data fact

`R9_condition_df.rds` was pulled with concept sets for depression, anxiety, diabetes, hypertension, COPD, obstructive sleep apnea, heart failure, myocardial infarction, stroke, and IBS (concept `75576`)—but *not* GERD. GERD must therefore be ascertained through a dedicated Cohort Builder export; it cannot be recovered from the existing condition table. Peptic ulcer disease is likewise absent and is supplied via the pre-computed `R9_pud_status.rds`.

6.6 Verified schemas: raw tables

Every column name and type used by the analysis code was confirmed by opening the dataset schema definitions directly, rather than inferred from the antecedent scripts. This proved essential: several assumptions that looked entirely reasonable turned out to be wrong, and are documented in Section 6.8 and Section 33.4. Table 6 records the raw inputs, their scale, and the specific columns the pipeline consumes.

Table	Rows	RAM	Columns consumed
`R9_fitbit_sleep_daily_summary_df`	31.8M	2.4 GB	`sleep_date` (Date), `is_main_sleep` (chr), `minute_in_bed`, `minute_asleep`, `minute_awake`, `minute_restless`, `minute_deep`, `minute_light`, `minute_rem`
`R9_fitbit_activity_df`	44.2M	2.4 GB	`date` (Date), `fairly_active_minutes`, `lightly_active_minutes`, `sedentary_minutes`, `very_active_minutes`
`R9_fitbit_intraday_steps_df`	34.2M	1.0 GB	`date` (Date), `wear_hours`, `total_steps`
`R9_drug_df`	14.5M	**3.1 GB**	`drug_class_concept_id`, `drug_exposure_start_datetime` (chr)
`R9_survey_df`	9.7M	0.7 GB	`question_concept_id`, `answer_concept_id`, `survey_datetime` (chr)
`R9_condition_df`	1.6M	274 MB	`condition_concept_id`, `condition_start_datetime` (chr), `standard_concept_name`
`R9_measurement_df`	50,942	14 MB	`measurement_concept_id`, `value_as_number`, `measurement_datetime` (chr)
`R9_person_df`	59,018	5 MB	`date_of_birth` (**chr**), `race`, `ethnicity`, `sex_at_birth`, `gender`

Table 6: Raw inputs with verified columns. Note the type asymmetry: EHR and survey date-like fields are stored as *character* and require explicit `as.Date()` conversion, whereas the Fitbit date fields are already `Date`. `person_id` is `numeric` in every table, so joins are type-consistent throughout.

Two facts in Table 6 shape the architecture. First, `R9_condition_df` contains the comorbidity concepts and IBS but *not* GERD, so a dedicated outcome pull is unavoidable. Second, the memory column: loading the sleep, activity, intraday-steps and drug tables simultaneously approaches nine gigabytes before any analysis begins. A pipeline that naively re-derives every covariate from raw data risks exhausting the workspace, which directly motivates the reuse strategy of Section 14.5.

6.7 Verified schemas: pre-built covariate objects

The project already contains a substantial library of derived, outcome-agnostic objects produced by the published IBS and constipation pipelines. Because these are the very objects underpinning the published analyses, they are the preferred inputs for the GERD study.

Object	Rows	Contents
Sleep-anchored		
`valid_population_sleep`	**19,995**	**The published cohort.** `shared_fitbit`, `shared_ehr`, `age_at_fitbit_start`, `age_cat`
`sleep_summary_filtered`	23,812	`n_valid_nights` plus the eight `avg_*` sleep metrics
`bmi_covariates_sleep_df`	46,753	`median_bmi`, `mean_bmi`, `bmi_closest`, and `first_fitbit_sleep_date`
`cci_covariates_sleep_df`	11,652	`cci_score` (int), `cci_cat` (factor)
`medication_flags_sleep`	11,105	the seven `on_*` flags
`comorbidity_status_sleep_ibs_df`	8,079	nine `has_*` indicators (*no* `has_pud`)
`age_at_fitbit_sleep_df`	54,180	`age_at_fitbit_start`, `age_cat`
Activity-anchored		
`R9_valid_population`	43,966	activity-cohort eligibility and age
`R9_steps_summary_filtered`	50,026	`avg_daily_steps`, `n_valid_days`
`R9_activity_zone_summary`	55,552	zone minutes, including a ready-made `avg_total_active_min`, plus `n_days_activity`
`R9_avg_wear_hours_df`	55,557	`avg_daily_wear_hours`, `n_valid_days`
`R9_comorbidity_status_df`	8,626	nine `has_*` *plus* `has_pud`, `has_gastroenteritis`, `has_h_pylori`
`R9_bmi_covariates_activity_df`	49,629	as the sleep version, keyed on `first_fitbit_date`
`R9_max_hr_minute_all_days_df`	53,812	`avg_daily_max_hr_minute_all_days` and quantiles
Shared		
`R9_pud_status`	**137**	`has_pud`. Extremely rare — see Section 25
`cci_scored_sleep`	11,652	raw `cci_score` before categorisation

Table 7: Pre-built covariate objects with verified row counts and contents. That `valid_population_sleep` contains exactly 19,995 participants — the published IBS paper's analytic N — is the strongest available check that the GERD analysis is operating on the same cohort.

6.8 Join hazards in the schema

Four features of these schemas silently corrupt naive joins. Each is handled explicitly in the code, and each is worth recording because they are invisible in the antecedent scripts.

1. **Duplicated `n_valid_days`.** Both `R9_steps_summary_filtered` and `R9_avg_wear_hours_df` carry a column of that name. Joining them yields `n_valid_days.x` and `n_valid_days.y`, so every later reference to `n_valid_days` — including the descriptive table's coverage row — fails or silently selects the wrong column. The wear-hours object is therefore reduced to `person_id` and `avg_daily_wear_hours` before joining.
2. **Suffixed survey columns.** `alcohol_summary_df` was itself built by joining three survey extracts and carries `answer_concept_id.x` and `answer_concept_id.y`. These contaminate the covariate bundle and collide with the education and income extracts, which also carry `answer_concept_id`. Only `alcohol_likert_final` is taken.

3. **Free-text socio-economic values.** `education_df` and `income_df` carry both `answer_concept_id` and a free-text `*_response`. The antecedent IBS script matched on the text, and its own comments record the consequence: the observed response strings did not match its `case_when` arms, so essentially every participant collapsed into "Missing". The GERD code therefore maps categories from the *concept identifiers*, which are stable, with the text matching retained only as a fallback.

4. **En-dashed factor levels.** `age_cat` uses levels `<30`, `30–44`, `45–59`, `60–74`, `75+` and `cci_cat` uses `0`, `1–2`, `3–4`, `5+`, all with typographic en-dashes rather than hyphens. Code written with hyphens fails to match, producing either silent `NA`s or a stream of level-mismatch warnings. Dash style is normalised before the factors are used.

These are not hypothetical concerns; items 1, 2, 3 and 4 all appeared as concrete defects during development and are catalogued in Section 33.4.

7 Outcome Ascertainment: GERD and Oesophagitis

7.1 Two outcomes

The study models two binary outcomes, each ascertained from the electronic health record and each requiring at least two condition records.

Outcome	Source	Rule
`has_gerd_any`	seed **318800** (SNOMED 235595009) + descendants	$\geq$ 2 records
`has_esophagitis`	seed **30753** (SNOMED 16761005) + descendants	$\geq$ 2 records

Erosive oesophagitis (**4231067**) is folded into the oesophagitis concept set: it is the same phenotype coded with more specificity, and excluding it would place those participants in the "no oesophagitis" group. The two outcomes are not mutually exclusive — a participant may carry codes for both, and that overlap is reported descriptively rather than being made into a third outcome.

The six analysis configurations. Two outcomes crossed with three exposure sets:

Exposure set	`has_gerd_any`	`has_esophagitis`
(A) Sleep quartiles	Config 1	Config 2
(B) Activity quartiles	Config 3	Config 4
(C) Combined (+ mutually adjusted)	Config 5	Config 6

Each configuration produces Table 1, Table 2, two supplementary tables, two figures and a diagnostics summary. The spline analysis of restricted cubic splines and the head-to-head comparison, specified later in this report, run over the same six.

What is retained descriptively. `build_gerd_outcomes()` derives a non-oesophagitic flag and the count of participants carrying any oesophagitis record, and prints both under a DESCRIPTIVE (NOT MODELLED) heading. The GERD/oesophagitis overlap is a number the manuscript reports, and it should come from the same code path that defines the outcomes rather than a separate tabulation. Nothing about it enters a model, a table, or the multiplicity correction.

Multiplicity. Bonferroni correction is applied within outcome, across exposures.

7.2 The descendant-filter subtlety

Each pull is *already* restricted to its own descendant-expanded concept set, so each phenotype is computed over *all* rows of its own file rather than by re-filtering on the seed id. Re-filtering on only `c(318800)` or `c(30753)` would silently drop every descendant-coded case and badly undercount the phenotype. Membership is therefore determined purely by which file a record came from.

7.3 Case-definition variants

Variable	Rule	Purpose
`has_gerd_any`	$\geq$ 2 records in the 318800 pull, first $\geq$ 180 days after the first Fitbit record	Primary GERD outcome (paper-matching temporal rule; `GERD_PRIMARY_DEF` switches to the "ever" rule).
`has_esophagitis`	$\geq$ 2 records in the 30753 pull	Primary erosive outcome.
`has_gerd_any_sens`	the alternative case definition	Sensitivity analysis.
`has_eso_any_record`	$\geq$ 1 oesophagitis record	Descriptive only: quantifies the overlap between the two outcomes.
`has_gerd_no_eso`	GERD case with no oesophagitis record	Descriptive only; not modelled (see above).
`severe_gerd_binary`	GERD *and* $\geq$ 2 post-diagnosis PPI/H_2RA dates	Optional medication-treated (severe) GERD.

Table 8: Outcome and case-definition variants. Only the first two are modelled.

8 Concept-Set Construction in Depth

Because outcome and drug-class definitions rest entirely on the correctness of the concept sets, this section walks through the two Cohort Builder mechanisms the pulls rely on.

8.1 Condition descendant expansion (`cb_criteria`)

The outcome pull does not enumerate GERD concept IDs by hand; it seeds two standard concepts and lets BigQuery expand the hierarchy. The nested query has three logical steps:

1. **Seed to hierarchy rows.** A subquery selects rows of `cb_criteria` whose `concept_id` is the seed (4144111 or 30753) and whose `full_text` marks them as standard "rank-1" entries, returning their hierarchy `id`s.
2. **Path match to descendants.** The outer `cb_criteria` scan keeps every standard, selectable concept whose materialized `path` lies on, above, or below a seed id — captured by the four `LIKE`/equality patterns on `path` — which is exactly the seed's descendant subtree in the SNOMED hierarchy.
3. **Restrict to condition occurrences for Fitbit-sharers.** The resulting concept set filters `condition_occurrence`, and a `cb_search_person` subquery restricts to participants with `has_fitbit = 1`.

This is why re-filtering the pulled table on only the two seed IDs (Section 7.2) is wrong: the pulled rows carry *descendant* concept IDs, and the two seeds are only the roots of the subtree.

8.2 Drug ancestor rollup (`concept_ancestor`)

The pharmacologics pull uses the complementary mechanism. A double join through `concept_ancestor` maps each `drug_exposure` to its ancestor drug classes, and the query keeps rows whose `ancestor_concept_id` is a target class (PPI `21600095` or H_2RA `21600096`). This rolls individual products (ingredients, brands, formulations) up to the class level, so the treated-GERD flag responds to any PPI or H_2RA regardless of the specific product coded.

8.3 Verification workflow

Both concept sets are seeds to be confirmed in the Cohort Builder before a production run. The outcome upload prints a `count()` of the captured concepts and the size of the name-matched oesophagitis subset, so the analyst can confirm the definition matches the intended clinical scope. If additional reflux/erosive-esophagitis concepts sit outside the two GERD seeds' subtree, their seed IDs are added to the `concept_id IN (...)` list; the name-match rule for `has_esophagitis` then picks them up automatically.

9 Exposure Definitions

9.1 Sleep metrics

For each retained night the pipeline keeps minutes asleep, in bed, awake, restless, and in the deep, light, and REM stages. Sleep efficiency is derived per night as

$$\text{efficiency} = \frac{\text{minutes asleep}}{\text{minutes in bed}}, \qquad \text{defined only when minutes in bed} > 0.$$

Nightly values are averaged within participant to yield stable exposures (`avg_min_asleep`, `avg_min_in_bed`, `avg_min_awake`, `avg_min_restless`, `avg_min_deep`, `avg_min_light`, `avg_min_rem`, `avg_sleep_efficiency`) with `n_valid_nights`. Each averaged metric is discretized into cohort quartiles with `ntile(..., 4)` and stored as a four-level factor with **Q1 as reference**. Modeling quartiles rather than raw minutes sidesteps linearity assumptions, is robust to outliers, and yields interpretable dose–response patterns. The sleep-cleaning cascade retains main sleep only, keeps nights with minutes asleep in $(0, 1440]$, excludes participants with $\geq 30\%$ of nights below 240 minutes (4 hours), and requires ≥ 180 valid nights.

9.2 Activity metrics

A *valid activity day* has ≥ 10 wear hours and between 100 and 45,000 steps (pre-filtered in the intraday-steps table); participants need ≥ 30 valid days. On those days the pipeline computes average daily steps (`avg_daily_steps`), activity-zone minutes (`avg_lightly_active_min`, `avg_fairly_active_min`, `avg_very_active_min`, `avg_total_active_min`, `avg_sedentary_min`), average daily wear hours, and, when available, the corrected maximum daily heart rate (`avg_daily_max_hr_minute_all_days`). Each is quartilized (Q1 reference); the max-HR exposure is guarded and omitted if its pre-built summary is absent.

9.3 Sleep×activity overlap sensitivity set

A stricter sensitivity population retains only sleep-nights that coincide with a valid activity day (a same-day intraday-steps record), yielding `sleep_activity_overlap_df` and a parallel summary. This isolates participants with concurrently well-measured sleep and activity.

Exposure set	**Metric**	**Quartile variable**
Sleep (8)	minutes asleep / in bed / awake / restless; deep; light; REM; efficiency	`min_asleep_quartile`, `min_in_bed_quartile`, `min_awake_quartile`, `min_restless_quartile`, `min_deep_quartile`, `min_light_quartile`, `min_rem_quartile`, `sleep_efficiency_quartile`
Activity (7)	daily steps; lightly/fairly/very/total active minutes; sedentary minutes; corrected max HR	`step_quartile`, `lightly_active_quartile`, `fairly_active_quartile`, `very_active_quartile`, `total_active_quartile`, `sedentary_quartile`, `max_hr_minute_all_days_quartile`

Table 9: The fifteen quartile exposures. The combined analysis models all fifteen (each separately and in mutually-adjusted pairs).

10 Fitbit Measurement and Reliability

The credibility of an objectively-measured exposure rests on understanding how the device produces each metric and where it is fallible. This section documents the provenance of the sleep and activity streams and the corresponding cleaning choices.

10.1 Sleep staging

Fitbit estimates sleep stages (light, deep, REM) and wake from movement (accelerometry) and photoplethysmography-derived heart-rate variability during a main sleep episode. The device reports, per night, minutes asleep, minutes in bed, minutes awake, minutes restless, and minutes in each stage. Stage estimates are less reliable than the coarser asleep/awake distinction, which is why the analysis models each stage *separately* as a quartile and does not, for example, sum stages into a single composite that would propagate stage-classification error. Nights of non-main sleep (naps) are excluded because their architecture differs systematically from nocturnal sleep. The $(0, 1440]$-minute validity window removes records that are physiologically impossible and typically reflect device or sync errors. Averaging over ≥ 180 nights suppresses night-to-night noise so that each participant's quartile placement reflects a habitual pattern rather than a few unusual nights.

10.2 Sleep efficiency

Sleep efficiency, minutes asleep divided by minutes in bed, is a unitless summary of sleep continuity that is more robust than any single stage because it depends only on the well-measured asleep/in-bed quantities. It is defined only when minutes in bed is positive, avoiding division-by-zero artifacts.

10.3 Steps, wear time, and valid days

Daily steps come from the intraday-steps stream, which also yields an estimate of *wear hours* per day. A day is counted as valid only when wear time is ≥ 10 hours and the step total lies between 100 and 45,000; the lower step bound removes non-wear or malfunction days that would masquerade as extreme sedentariness, and the upper bound removes implausible sensor spikes. Requiring ≥ 30 valid days yields a stable habitual activity profile while retaining a large sample.

10.4 Activity-zone minutes

The daily activity stream partitions each day into sedentary, lightly active, fairly active, and very active minutes based on metabolic-equivalent thresholds inferred from movement and heart rate. The analysis models each zone as its own quartile and additionally forms a total-active-minutes quartile (the sum of the three active zones), so that both the *intensity distribution* and the *overall active volume* are represented. Zone minutes are computed only on valid days, so that a low active-minute count reflects genuine inactivity rather than device non-wear.

10.5 Corrected maximum heart rate

The corrected maximum daily heart rate is drawn from a pre-computed summary (`R9_max_hr_-minute_all_days_df.rds`) that addresses artifacts in the raw minute-level maxima. It serves as a cardiovascular-fitness and autonomic proxy in the activity set. Because that summary is a separate artifact, the exposure is guarded: if the file is absent, the max-HR quartile is simply omitted and the remaining activity exposures proceed unaffected.

10.6 Implications for interpretation

Two measurement facts shape interpretation. First, wearable exposures are *habitual averages*, so the estimands concern long-run typical sleep and activity, not acute perturbations. Second, exposure measurement error, being device-based and plausibly non-differential with respect to EHR-coded reflux, would tend to bias quartile contrasts toward the null; observed associations are therefore likely to be conservative.

11 Combined Exposure Design

11.1 Intersection cohort

The combined analysis is defined on participants who satisfy *both* exposure-eligibility thresholds: ≥ 180 valid sleep nights *and* ≥ 30 valid activity days, plus EHR-sharing and age ≥ 18. This is a genuine intersection (an `inner_join` of the two eligible sets), so its sample is smaller than either single-exposure cohort but every participant contributes both a complete sleep profile and a complete activity profile.

11.2 Covariate-timing anchor

For the single-exposure analyses, covariate windows are anchored to the first Fitbit date of the relevant stream (first sleep date for the sleep file; first activity date for the activity file). For the combined analysis, a participant may have started the two streams on different days, so covariate timing is anchored to the *earliest* Fitbit date across streams,

$$\texttt{first\_fitbit\_date} = \min(\texttt{first\_sleep\_date}, \texttt{first\_activity\_date}),$$

computed element-wise with `pmin(..., na.rm = TRUE)`. All pre-Fitbit covariate windows (co-morbidities, medications, BMI) are then defined consistently regardless of which stream began first.

11.3 Two model families

For each outcome the combined file fits: **(A)** one adjusted model per exposure, for all fifteen exposures, via the shared engine; and **(B)** *mutually-adjusted* models that include one sleep quartile and one activity quartile simultaneously, so each exposure is adjusted for the other domain in addition to the full covariate set. For a sleep exposure S, an activity exposure A, and covariates $\mathbf{Z}$,

$$\operatorname{logit}\big(\Pr[Y=1]\big) = \beta_0 + \sum_{j=2}^{4} \beta_j^S \mathbb{1}[S{=}Q_j] + \sum_{j=2}^{4} \beta_j^A \mathbb{1}[A{=}Q_j] + \sum_k \gamma_k Z_k.$$

Four primary sleep×activity pairs are examined: minutes-asleep×steps, efficiency×very-active, deep-sleep×total-active, and REM×sedentary. These pairings probe whether a sleep association survives adjustment for activity and vice versa.

12 Cohort Construction and Eligibility

12.1 Eligibility gates

Three participant-level gates apply to every configuration: age ≥ 18 at the anchoring first-Fitbit date; shared EHR data (present in the condition *or* measurement table); and shared Fitbit data, implied by meeting the stream-specific validity threshold (≥ 180 sleep nights, ≥ 30 activity days, or both for the combined cohort). Sleep cleaning additionally applies the main-sleep, valid-minute, and short-sleep rules of the previous section.

12.2 Clinical exclusions

(iii) Foregut surgery and upper gastrointestinal cancer are exclusions. A myotomy deliberately cuts the lower oesophageal sphincter, so reflux afterwards is produced *by the operation*. Fundoplication, gastrectomy, bariatric surgery and oesophagectomy likewise change the anatomy that generates reflux. Oesophageal and gastric cancer alter eating, weight and sleep, produce reflux-like symptoms, and lead to major surgery. Neither group is comparable to the population this study is about, so both are removed rather than adjusted for. Achalasia is retained as an exclusion: it mimics reflux and is the indication for the myotomies above.

Oesophageal dilation and bougienage are deliberately *not* excluded. They treat a stricture, which is a consequence of reflux rather than a cause; excluding them would remove severe-reflux participants and bias the association towards the null. They are available as covariates.

Concept ids exist for achalasia (318186), oesophageal malignancy (4181343), oesophagectomy (4187533) and gastrectomy (4203442). Heller myotomy, POEM and cricopharyngeal myotomy have no id in this workspace's concept set but appear by name in the procedure export, so exclusion matches on concept id where one exists and on name where it does not.

12.3 CONSORT accounting

Each analysis file reproduces a CONSORT-style waterfall counting unique participants surviving each step, from "any Fitbit record" through the final analytic cohort, recording the number excluded and the reason at each step.

Step	Sleep cohort criterion (kept)	Activity cohort criterion (kept)
0	Any Fitbit sleep record	Any Fitbit activity record
1	Main sleep & valid minutes (1–1440)	Valid day: ≥ 10 wear h, 100–45000 steps
2	$< 30\%$ nights with < 4h sleep	—
3	≥ 180 valid nights	≥ 30 valid days
4	Shared any EHR data	Shared any EHR data
5	Age ≥ 18 at first Fitbit	Age ≥ 18 at first Fitbit

Table 10: CONSORT steps for the sleep and activity cohorts. The combined cohort is the intersection of the two final rows.

13 Rationale for Eligibility and Cleaning Thresholds

Every numeric threshold in the pipeline is a deliberate trade-off between measurement quality and sample size, chosen to match the antecedent IBS and constipation studies so that results are comparable across the portfolio. Table 11 collects them.

Rule	Threshold	Rationale
Main sleep only	`is_main_sleep`	Naps have different architecture and would bias nightly stage means.
Valid sleep minutes	$(0, 1440]$	Zero or $>$ 24h asleep is physiologically impossible (device error).
Short-sleep exclusion	$<$ 30% nights $<$ 4h	Removes device-wear artifacts and extreme populations without discarding occasional short nights.
Sleep coverage	$\geq$ 180 nights	About six months of data stabilizes long-run nightly means; shorter windows are noisier.
Valid activity day	$\geq$ 10 wear h; 100–45000 steps	Ensures the device was worn most of the day and step counts are plausible (excludes non-wear and sensor spikes).
Activity coverage	$\geq$ 30 valid days	Balances a stable activity average against retaining enough participants.
BMI window	$\pm$180 days	Keeps adiposity contemporaneous with the exposure window.
Comorbidity rule	$\geq$ 2 distinct pre-Fitbit dates	Enforces temporality and specificity (single stray codes excluded).
Medication rule	$\geq$ 2 dates in prior 12 months	Captures sustained rather than incidental use.
Age gate	$\geq$ 18 at first Fitbit	Adult population; consistent denominator.

Table 11: Eligibility and cleaning thresholds and their justifications.

The sleep and activity coverage thresholds differ ($\geq$ 180 nights vs $\geq$ 30 days) because they inherit from two separate parent studies and because sleep and step streams have different missingness profiles; this asymmetry is why the combined cohort is a genuine intersection rather than a simple union, and why its sample is the smallest of the three.

14 Covariates

14.1 Temporal covariate rule

All EHR-derived comorbidities are ascertained *prior to* the anchoring first-Fitbit date and require $\geq$ 2 diagnoses on *distinct* dates, implemented by one helper, `create_covariate_status()`. Requiring two distinct dates before the exposure window enforces both temporality (the confounder precedes the exposure) and specificity (single stray codes do not qualify). This matches the functional-dyspepsia and constipation covariate builds, preserving cross-study consistency.

14.2 Covariate inventory

14.3 Medication, BMI, and GI covariate details

Medication "use" requires $\geq$ 2 distinct exposure dates within the 12 months preceding the first Fitbit date, capturing sustained rather than incidental use. BMI uses a within-person median (robust central tendency) and, in the single-exposure files, the single measurement closest to the first Fitbit date, preferring prior measurements and nulling values more than 180 days away. Two gastrointestinal covariates are added because they are clinically entangled with GERD: peptic

Domain	Variables	Source / definition
Demographics	age (+category), sex at birth, race, ethnicity	`R9_person_df`; collapsed.
Socio-economic	education, income	`R9_survey_df` items, collapsed.
Behavioral	smoking (ever $\geq$ 100 cigarettes), alcohol (Likert 0–5/day)	`R9_survey_df`.
Anthropometric	median BMI; nearest BMI within ±180 days	`R9_measurement_df` (concept 3038553).
Psychiatric / cardiometabolic	depression, anxiety, diabetes, hypertension, heart failure, MI, stroke, COPD, sleep apnea	`R9_condition_df` via `create_covariate_status()`.
Comorbidity burden	Charlson Comorbidity Index (continuous & categorical)	`cci_scored_sleep.rds` / `R9_cci_scored.rds`.
Medications	beta-blockers, calcium-channel blockers, stimulants, antidepressants, antipsychotics, anxiolytics, hypnotics	`R9_drug_df`; $\geq$ 2 dates within 12 months before first Fitbit.
GI (GERD-specific)	peptic ulcer disease (`has_pud`)	From the condition export via concept 4027663.

Table 12: Covariate inventory. The final row is the GERD-specific addition relative to the IBS study.

Condition	Concept ID	Condition	Concept ID
Depression	440383	Stroke	381316
Anxiety	441542	COPD	255573
Diabetes	201820	Sleep apnea (OSA)	313459
Hypertension	316866	IBS (covariate)	75576, 4234788, 4261072, 4057826
Heart failure	316139	Peptic ulcer disease	4318534, 4134146, 4265600, 4027663
Myocardial infarction	4329847		

Table 13: Condition concept IDs used to build covariates.

ulcer disease (`has_pud`, shared acid pathophysiology and PPI treatment) from `R9_pud_status.rds`, and IBS (`has_ibs`), rebuilt *as a covariate* from `R9_condition_df` using the same pre-Fitbit, $\geq$ 2 distinct-dates rule so that GERD and IBS can co-occur while the temporal rule stays consistent.

14.4 Confounding control on GERD expert review

A GERD expert reviewed the protocol and required three changes. All are implemented in `GERD Concepts R9.R` and `GERD Analysis Helpers R9.R`.

(i) Sleep medication is now an adjustment variable. Tricyclic antidepressants (amitriptyline, nortriptyline, doxepin and related) and hypnotics — both benzodiazepine (triazolam/Halcion, temazepam) and non-benzodiazepine (zolpidem, eszopiclone) — alter sleep duration and architecture *and* several reduce lower oesophageal sphincter tone. Omitting them attributes a pharmacological effect to sleep itself. The covariate `on_sleep_med` is the union of the three classes; the components are retained so the composite is auditable and any one class can be modelled alone.

(ii) Obstructive sleep apnoea is now an adjustment variable. This is the dominant confounder of any sleep–reflux association. Apnoea causes both fragmented sleep and reflux, and it travels with obesity. An unadjusted estimate of "poor sleep $\rightarrow$ GERD" partly measures undiagnosed apnoea instead. `has_sleep_apnea` (concept 313459) enters every adjusted model.

(ii-b) Opioid analgesics are adjusted for. Added when the study team exported a dedicated `Narcotics` dataset. Opioids delay gastric emptying and reduce lower oesophageal sphincter tone, and they fragment sleep — they confound both sides of the association, so leaving them out would load an opioid effect onto the sleep coefficient.

The adjustment set, as revised.

$$\begin{aligned}\text{logit}\Pr(\text{outcome}) = \beta_0 &+ \beta_1\text{Exposure quartile} + \gamma_1\text{age} + \gamma_2\text{sex} + \gamma_3\text{race} + \gamma_4\text{ethnicity}\\ &+ \gamma_5\text{education} + \gamma_6\text{income} + \gamma_7\text{smoking} + \gamma_8\text{alcohol} + \gamma_9\text{BMI}\\ &+ \gamma_{10}\,\textbf{sleep apnoea} + \gamma_{11}\,\textbf{sleep medication}\\ &+ \gamma_{13}\text{depression} + \gamma_{14}\text{anxiety} + \gamma_{15}\text{diabetes} + \gamma_{16}\text{hypertension} + \gamma_{17}\text{peptic ulcer diseas}\end{aligned}$$

Two departures from the published IBS template. `has_ibs` is dropped: IBS is not in this workspace's concept set, so it was constant and contributed nothing. The comorbidity index is dropped from the default set: it is built from congestive heart failure, COPD, type 2 diabetes and peptic ulcer disease, so including it alongside `has_diabetes` and `has_pud` double-counts them. Diabetes and hypertension are adjusted for explicitly instead, which is both what the expert asked for and more interpretable. Setting `GERD_USE_COMORBIDITY_INDEX <- TRUE` restores the index in place of the explicit flags. Note that the index available here is a *partial* count over the four Charlson conditions present in this concept set — it is not a validated Charlson Comorbidity Index and is labelled accordingly in every table.

Participant flow. The build writes `participant_flow.csv`: the count at every step from "has Fitbit data" through the exposure minimum, EHR sharing, the age threshold and the clinical exclusions, for both the sleep and activity cohorts. The manuscript's flow figure is transcribed from that file rather than reconstructed after the fact.

14.5 Covariate provenance: reuse versus re-derivation

A decision that materially affects both the statistics and the practicality of this study is where the covariates come from. Two options exist: re-derive every covariate from the raw tables inside the GERD pipeline, or reuse the derived objects already present from the published IBS and constipation work (Table 7).

The pipeline **prefers the pre-built objects**, falling back to re-derivation only when one is absent. This is a methodological choice before it is an engineering one, and it rests on three arguments.

Comparability. The covariate derivations are outcome-agnostic: whether a participant had two distinct pre-Fitbit depression codes has nothing to do with whether they later develop reflux. Those objects already underpin the published IBS paper. Reusing them guarantees that any difference between the GERD and IBS results reflects the outcome rather than incidental divergence in how a covariate was recomputed — a subtle but real threat when two pipelines implement "the same" rule independently.

Feasibility. Re-deriving medication flags requires `R9_drug_df`, approximately 3.1 GB in memory; re-deriving the sleep summary requires the 31.8M-row nightly table at 2.4 GB. A pipeline that loads several of these at once approaches nine gigabytes, which is a realistic cause of failure in a shared workspace. Reuse removes that risk entirely for the common case.

Transparency. Because reuse could in principle mask a stale or mismatched input, the loader announces its choice for every covariate block, printing a `[load]` line naming the file it used or a `[miss]` line listing the candidates it could not find before falling back. The provenance of every covariate in the final frame is therefore visible in the run log rather than implicit.

The one covariate that is always rebuilt rather than reused is `has_ibs`, which is retained in the data layer but is *not* in the adjustment set: IBS is absent from this workspace's concept set, so the flag was constant and contributed nothing. The available `ibs_status` objects encode the IBS *outcome* definition (at least two records at any time), whereas as a covariate IBS must follow the same pre-Fitbit, two-distinct-dates rule as the other comorbidities. The pipeline therefore derives it from `R9_condition_df`, loading that table only for this purpose and releasing it immediately afterwards, and falls back to the pre-built status object with an explicit warning only if the condition table is unavailable.

14.6 A caveat on covariate anchoring

Covariates are defined relative to a participant's first Fitbit record, but "first Fitbit record" differs between the sleep and activity streams. The sleep analysis uses sleep-anchored objects, the activity analysis activity-anchored ones. Two consequences follow and are stated plainly rather than hidden. First, `has_pud` is only available from an object computed on the activity anchor, so in the sleep analysis it carries a small timing inconsistency relative to the other comorbidities. Second, the combined cohort has two candidate anchors; it uses the sleep anchor by default, on the grounds that every member of that cohort has at least 180 valid sleep nights so the anchor is always well defined, with a single configuration switch to use the activity anchor instead.

15 Covariate-by-Covariate Rationale

Each covariate earns its place by lying on a backdoor path between the wearable exposures and reflux. This section records the specific justification for each, so the adjustment set can be scrutinized individually rather than as a block.

- **Age.** Reflux prevalence and coding rise with age, and sleep architecture and activity both change across the lifespan; modeled as a five-level category to allow non-linear age effects.
- **Sex at birth.** Sex differences exist in reflux presentation, erosive-disease risk, and in sleep and activity patterns.
- **Race and ethnicity.** Correlate with both health-care access/coding and behavioral patterns; collapsed to protect small cells while retaining the major contrasts.
- **Education and income.** Socio-economic status shapes device access, health-care engagement (hence diagnosis probability), and lifestyle; both are carried with an explicit Unknown/Missing level.
- **Smoking.** Nicotine lowers lower-esophageal-sphincter tone (a direct reflux mechanism) and degrades sleep; a classic confounder of any sleep–reflux association.
- **Alcohol.** Relaxes the sphincter, disrupts sleep architecture, and associates with activity; modeled on the daily-quantity Likert scale.
- **Median BMI.** The single most important confounder: adiposity raises intra-abdominal pressure and promotes reflux and hiatal herniation while also shortening sleep and reducing activity.
- **Charlson Comorbidity Index.** Captures residual multimorbidity that drives both health-care contact (and thus coding) and reduced sleep/activity; categorical by default.
- **Depression and anxiety.** Alter sleep, elevate visceral symptom reporting, and associate with reflux diagnosis; adjusted directly rather than folded into the Charlson index.
- **Peptic ulcer disease.** Shares acid pathophysiology and PPI treatment with GERD; a GERD- specific covariate absent from the antecedent IBS study.

- **Irritable bowel syndrome.** Frequently overlaps GERD and disturbs sleep; entered as a covariate (not the outcome), rebuilt from the condition table under the same temporal rule.
- **Diabetes, hypertension, heart failure, MI, stroke, COPD, sleep apnea.** Cardiometabolic and respiratory comorbidities that pattern both physiology and health-care coding; sleep apnea in particular is strongly linked to both disturbed sleep and reflux.
- **Medication-use flags.** Beta-blockers, calcium-channel blockers, stimulants, antidepressants, antipsychotics, anxiolytics, and hypnotics all affect measured sleep and heart rate and may mark conditions linked to reflux; each is a $\geq$ 2-date pre-exposure flag.

Acid-suppression therapy is deliberately *absent* from this list: as a probable mediator (downstream of reflux), it is used only to define the treated-GERD sensitivity outcome, never as an adjustment covariate.

16 Software Architecture

The implementation is deliberately layered so that all six configurations share one statistical engine and one data-loading path. Only three things differ between configurations: the cohort, the exposure list, and which outcome column the driver stratifies on. Everything else — estimator, covariate set, reference levels, missing-data handling, table format — is defined exactly once.

16.1 The four layers

File	Layer and responsibility
`GERD Outcome Data Upload R9.R`	**Acquisition.** BigQuery pull of GERD condition records with Cohort Builder descendant expansion, previewing both the captured concepts and the oesophagitis subset. Produces `R9_gerd_outcome.rds`. Run once.
`GERD Data Prep R9.R`	**Data layer.** Schema-aware loading of cohort, exposures and the one-row-per-participant covariate bundle. Implements prefer-pre-built-else-derive, concept-ID mapping for socio-economic variables, NULL-safe joins, column slimming against the join hazards, and memory release after large tables are reduced.
`GERD Analysis Helpers R9.R`	**Engine.** Outcome definitions and labels, exposure lists, adjustment set, master label lookup; quartile cutoff labelling; descriptive, univariable and multivariable routines; the separation guard; assumption diagnostics; and the manuscript tables and figures.
`Sleep and GERD Analysis File R9.R`	**Driver.** Configurations C1 and C2.
`Activity and GERD Analysis File R9.R`	**Driver.** Configurations C3 and C4.
`Activity Sleep and GERD Analysis File R9.R`	**Driver.** Configurations C5 and C6, plus the mutually-adjusted sleep×activity pair models.
`GERD Pharmacologics Data Upload.R`	**Optional acquisition.** PPI (21600095) and H_2RA (21600096) exposures for the treated-GERD sensitivity analysis.

Table 14: Code layers. Each driver sources the data layer and the engine, then delegates.

16.2 What a driver actually contains

Because the two shared files carry the logic, a driver reduces to a short, readable sequence: load the outcome source, build both phenotypes, load covariates, assemble one row per participant, create integer quartiles, and hand off to the engine.

```r
source("GERD Analysis Helpers R9.R")   # engine
source("GERD Data Prep R9.R")          # data layer

R9_gerd_outcome        <- read_first_existing("R9_gerd_outcome.rds", required = TRUE)
R9_gerd_outcome_status <- build_gerd_outcomes(R9_gerd_outcome,
                             first_fitbit_sleep_date_df, "first_fitbit_sleep_date")
covars_df              <- gerd_load_covariates(anchor = "sleep",
                             fitbit_dates = first_fitbit_sleep_date_df,
                             date_col = "first_fitbit_sleep_date")

final_analysis_sleep_gerd_df <- valid_population_sleep %>%
  inner_join(sleep_summary_filtered, by = "person_id") %>%
  left_join(covars_df, by = "person_id") %>%
  left_join(R9_gerd_outcome_status, by = "person_id") %>%
  apply_primary_outcome_def() %>%          # selects the primary case definition
  mutate(min_asleep_quartile = ntile(avg_min_asleep, 4), ...)

modeling_df   <- prep_modeling_df(final_analysis_sleep_gerd_df, SLEEP_EXPOSURES)
sleep_results <- analyze_all_outcomes(modeling_df, exposures = SLEEP_EXPOSURES,
                                      coverage_var = "n_valid_nights", stub = "sleep")
```

Listing 1: The essential structure of every driver.

16.3 The data layer in detail

`GERD Data Prep R9.R` exposes a small number of functions with clearly separated concerns.

- `read_first_existing(candidates, label, required)` — loads the first file that exists from a candidate vector, logging which one it used. This is the mechanism behind prefer-pre-built-else-derive, and it makes provenance explicit in the run log.
- `build_gerd_outcomes(pull, fitbit_dates, date_col)` — derives all four outcome columns (`has_gerd_any_ever`, `has_gerd_any_post_fitbit`, and the two oesophagitis equivalents) from the single pull, and prints the case counts under each rule.
- `gerd_load_covariates(anchor, ...)` — assembles the covariate bundle for the sleep or activity anchor, applying the column-slimming rules of Section 6.8 and coalescing binary EHR indicators to `FALSE`.
- `map_education()` / `map_income()` — concept-ID mapping with text fallback.
- `derive_comorbidities()`, `derive_medications()`, `derive_bmi()`, `derive_smoking()`, `derive_alcohol()`, `derive_ibs_covariate()` — the re-derivation fallbacks, used only when a pre-built object is missing.
- `drop_big()` — removes named large objects and triggers garbage collection, used after each multi-gigabyte table has been reduced to its summary.

Two defensive details are worth noting. Joins in the covariate assembler use a NULL-safe wrapper, so a missing covariate block degrades to "that covariate is absent" rather than aborting the run. And the assembler guarantees that `has_pud` and `has_ibs` exist in the output even if no source supplied them, setting them to `FALSE` with an explicit message — which the separation guard will then correctly identify as single-level and drop from the models.

16.4 The engine in detail

`GERD Analysis Helpers R9.R` holds every modelling decision in one place.

- **Configuration constants** — `GERD_PRIMARY_DEF` (default `"post_fitbit"`), `GERD_POST_FITBIT_LAG_DAYS` (180), `GERD_P_ADJUST` (`"bonferroni"`), `GERD_MIN_CELL` (1), `GERD_OUTPUT_DIR`, and the descriptive test choices.
- `GERD_OUTCOMES` — the two outcomes and their factor labels, driving the both-outcome loop.
- `SLEEP_EXPOSURES`, `ACTIVITY_EXPOSURES`, `GERD_ADJ_COVARS`, `GERD_LABELS`, and the exposure-to-source and exposure-to-unit maps used by Table 2.
- `apply_primary_outcome_def()` — selects which case definition becomes primary and retains the other as `*_sens`.
- `prep_modeling_df()` — builds collapsed factor covariates and relevels quartiles to Q1–Q4. Deliberately *defensive*: each collapsed variable is constructed only if it is not already present and its source column exists, so the same function works whether the frame arrived from pre-built objects (which already supply `education_collapsed`) or from re-derivation.
- `make_quartile_labels()` — computes the numeric boundaries and renders the `Q1 (<368)` style labels required by the target Table 2, with per-metric scaling (sleep efficiency is reported as a percentage) and thousands separators for step counts.
- `drop_unstable_covariates()` — the separation guard (Section 25).
- `run_models_for_outcome()`, `manuscript_table1/2()`, `manuscript_supp_univariate/multivariable()`, `forest_plot_exposures()`, `gvif_plot()`, `diagnostics_summary()`.
- `analyze_all_outcomes()` — the driver-facing entry point that loops both outcomes and emits the full manuscript set.

16.5 A note on package attachment

The engine installs but deliberately does *not* attach `MASS`, `car` and `pROC`, referencing them as `MASS::stepAIC`, `car::vif` and `pROC::roc`. The reason is concrete rather than stylistic: `MASS::select()` masks `dplyr::select()`, and because the engine is sourced before any data manipulation, attaching MASS breaks every subsequent `select()` call in the pipeline. The tidyverse packages are attached with `dplyr` last so its verbs take precedence. This defect was observed in practice (Section 33.4).

17 The Shared Modeling Engine and Descriptive Statistics

17.1 Engine functions

Every configuration is executed by `GERD Analysis Helpers R9.R`. Its exported objects and functions are:

- `GERD_OUTCOMES` — the two outcomes and their factor labels; `SLEEP_EXPOSURES` and `ACTIVITY_EXPOSURES`; `GERD_ADJ_COVARS`; and `GERD_LABELS` (master labels).
- `prep_modeling_df(df, exposures)` — builds collapsed factor covariates (race, ethnicity, sex, education, income, alcohol, smoking, Charlson category) and relevels each quartile exposure to Q1→Q4 with Q1 reference.
- `descriptive_table()`, `univariate_table()`, `run_models_for_outcome()`, `stack_results()`, `run_diagnostics()`.

- `analyze_all_outcomes(modeling_df, exposures)` — the driver that loops *both* outcomes, producing descriptive, univariable, multivariable, and diagnostic output for each.

Each analysis file builds its one-row-per-participant frame, creates integer quartiles, calls `prep_modeling_df()`, then `analyze_all_outcomes()`. Centralization guarantees that the sleep, activity, and combined analyses — and both outcomes within each — are estimated by identical code.

17.2 Descriptive statistics

For each outcome a `gtsummary::tbl_summary` table is stratified by the outcome (labeled "No GERD"/"GERD" or "No oesophagitis"/"Oesophagitis"), reporting the exposure quartiles, continuous exposure metrics, demographics, behavioral and socio-economic factors, the Charlson index, all comorbidities, PUD, IBS, medications, and median BMI. Chi-square tests are attached for the collapsed categorical demographics, Bonferroni-adjusted p-values are added, and emphasis is placed on q rather than raw p. Race, ethnicity, sex, education, and income are collapsed to protect small cells and improve legibility.

18 Reporting Conventions and Output Structure

Each configuration produces a fixed set of outputs in a fixed order, so that the six configurations can be read side by side. This section documents that structure.

18.1 Per-configuration outputs

For each outcome the engine emits, in order: (1) a stratified descriptive table (`tbl_summary`) with counts/percentages for categorical variables, medians/IQRs for continuous variables, a Missing row where applicable, chi-square tests for the collapsed demographics, and adjusted q-values; (2) a univariable table (`tbl_uvregression`) of unadjusted ORs with 95% CIs for every exposure and covariate; (3) a stacked multivariable table (`tbl_stack`) with, per exposure, a full model and a backward model side by side (`tbl_merge`), each footnoted with its complete-case N; and (4) a console diagnostics block. The combined file additionally emits the mutually-adjusted sleep$\times$activity tables.

18.2 Numbers and rounding

Odds ratios and confidence bounds are reported to two decimal places; percentages to one; the AUC to three. Cell sizes below the *All of Us* small-count reporting threshold must be suppressed or combined before any output leaves the workspace — a reason the demographic categories are collapsed in `prep_modeling_df()`.

18.3 What is fixed versus what varies

Across the six configurations the estimator, covariate set, reference levels, missing-data handling, and table format are *fixed* by the shared helper. Only three things *vary*: the cohort (which eligibility gate applies), the exposure list handed to the engine, and the outcome column the driver stratifies on. This separation is what licenses direct comparison of, say, the sleep effect on GERD (C1) with the sleep effect on oesophagitis (C2) or the activity effect on GERD (C3).

19 Manuscript Output Specification

The analysis does not emit exploratory objects for later reshaping; it emits the manuscript's tables and figures directly. Running any driver writes, for *both* outcomes, a fixed set of files into `./manuscript_output/`, prefixed by exposure set (`sleep_`, `activity_`, `combined_`) and then by outcome (`gerd_any_`, `esophagitis_`).

File	Manuscript element
`*_table1.csv`	**Table 1** — demographics and characteristics by outcome group, as median [IQR] for continuous variables and n (%) for categorical, with a single p-value column
`*_table2.csv`	**Table 2** — exposure metrics by outcome group, each participant-level average immediately followed by its quartiles, with the numeric cutoffs printed in the row labels
`*_supp_table1_univariate.csv`	**Supplement Table 1** — univariable logistic regression: N, OR, 95% CI, p, adjusted p; reference levels rendered as "–"
`*_supp_table2_multivariable.csv`	**Supplement Table 2** — multivariable models, exposure rows only, one model per metric, with per-model N
`*_figure1_forest.png`	**Figure 1** — forest plots of adjusted odds ratios, faceted by metric, Q1 as reference, log-scaled axis with a null line at 1
`*_figure2_gvif.png`	**Figure 2** — adjusted GVIF bar chart with the conventional threshold marked
`*_diagnostics.csv`	The numbers quoted in the Results prose: adjusted GVIF range, maximum Cook's distance, Box–Tidwell p-values for BMI and age, AUC range, duplicate-identifier count
`*_quartile_cutoffs.csv`	Exact unrounded quartile boundaries, for the Table 2 footnote stating that displayed cutoffs are rounded but exact values were used in analysis
`combined_*_mutually_adjusted_-pairs.csv`	Sleep×activity pair estimates (C5/C6)

Table 15: Emitted files and the manuscript elements they populate.

19.1 Quartile cutoff labelling

The target Table 2 does not merely report quartile membership; it reports the numeric boundary of each quartile in the row label, for example `Q2 (368-399)`. The engine computes these from the data rather than requiring them to be transcribed. Boundaries are taken from the empirical quartiles of the underlying continuous metric, rounded for display while the unrounded values are exported separately. Two per-metric refinements are applied: sleep efficiency is a proportion in the data but is reported as a percentage, so it is scaled before labelling; and step counts receive thousands separators.

```
Number of valid nights                    886 [532, 1,243]
Average sleeping minutes                    398 [369, 430]
Minutes asleep quartiles (minutes)
    Q1 (<369)                                    702 (25%)
    Q2 (369-398)                                 705 (25%)
    Q3 (399-430)                                 692 (25%)
    Q4 (>430)                                    693 (25%)
Average in bed minutes                      453 [420, 487]
Minutes in bed quartiles (minutes)
    Q1 (<421)                                    700 (25%)
    ...
Sleep efficiency                      0.885 [0.866, 0.905]
Sleep efficiency quartiles (%)
```

```
    Q2 (87-88)      Q3 (89-91)      Q4 (>91)
Daily steps quartiles (steps)
    Q1 (<5,785)     Q2 (5,785-7,489)     Q3 (7,490-9,108)     Q4 (>9,108)
```

Listing 2: Table 2 as produced by the code (excerpt, synthetic data).

The structure — coverage count, then for each metric its average followed by its labelled quartiles — reproduces the antecedent paper's Table 2 layout exactly, so the output can be transferred with formatting rather than restructuring.

19.2 Multiplicity in the emitted tables

Both supplement tables carry an adjusted p-value column alongside the raw one. The adjustment follows `GERD_P_ADJUST`, which defaults to Bonferroni across the models within an outcome, matching the antecedent paper. Setting it to `"fdr"` switches to Benjamini–Hochberg q-values without any other change.

20 Statistical Preliminaries and Notation

This section fixes the notation and the elementary quantities used throughout, so that later sections can refer to them without ambiguity.

20.1 Odds, odds ratios, and the logit

For a binary outcome $Y \in \{0,1\}$ with $p = \Pr[Y = 1]$, the *odds* are $p/(1-p)$ and the *logit* (log-odds) is $\operatorname{logit}(p) = \log\{p/(1-p)\}$, a monotone map from $(0,1)$ to $(-\infty,\infty)$ that makes a linear predictor sensible. The *odds ratio* (OR) comparing two covariate settings is the ratio of their odds; an OR of 1 denotes no association, < 1 lower odds, and > 1 higher odds. In a logistic model each coefficient β satisfies $e^{\beta} = \mathrm{OR}$ per unit change in its covariate, holding the others fixed; for a quartile contrast e^{β_j} is the OR of quartile j versus the Q1 reference.

20.2 The model and its likelihood

With covariate vector x_i and parameter vector $\theta = (\beta_0, \boldsymbol{\beta}, \boldsymbol{\gamma})$,

$$p_i(\theta) = \frac{1}{1 + \exp(-x_i^\top \theta)}, \qquad \ell(\theta) = \sum_{i=1}^{N} \{y_i \log p_i(\theta) + (1 - y_i)\log(1 - p_i(\theta))\}.$$

The maximum-likelihood estimate $\hat{\theta}$ maximizes ℓ; R's `glm` solves the score equations by iteratively reweighted least squares. Standard errors come from the inverse observed information, and a Wald 95% confidence interval for an OR is $\exp(\hat{\beta} \pm 1.96\,\widehat{\mathrm{SE}}(\hat{\beta}))$. Confidence intervals are reported on the OR scale throughout.

20.3 Discrimination, calibration, and fit

Model *discrimination* — the ability to rank cases above non-cases — is summarized by the area under the receiver-operating-characteristic curve (AUC, equivalently the concordance c-statistic), computed by `pROC`; 0.5 is chance and 1.0 is perfect. Because the outcomes are relatively rare, AUC is expected to be modest and is used comparatively (full vs backward model, across exposures) rather than as an absolute performance claim. Nested models are compared by AIC during backward selection. Separation and sparsity are monitored through coefficient standard errors, as described in the diagnostics.

20.4 Reference levels and contrasts

All quartile exposures use Q1 (lowest) as the reference; all categorical covariates use an explicit, clinically natural reference (`"<30"` for age, `"Non-Hispanic"` for ethnicity, `"0"` for the Charlson category, `"Non-smoker"` for smoking). Fixing references in one place — the shared helper's `prep_modeling_df()` — guarantees that every configuration reports contrasts against the same baseline, so ORs are comparable across the sleep, activity, and combined analyses.

21 Univariable and Multivariable Analysis

21.1 Univariable models

Each candidate predictor is regressed on the outcome individually via `gtsummary::tbl_uvregression` with `family = binomial` and exponentiated coefficients, yielding unadjusted odds ratios with 95% CIs for every exposure and covariate, including the GERD-specific `has_pud`, `has_sleep_apnea`, `on_sleep_med` and `on_narcotic` terms.

21.2 Multivariable models

Because the exposures within a set are collinear, a *separate* adjusted model is fit for each. For exposure X (a quartile factor with reference Q1) and covariate vector $\mathbf{Z}$,

$$\operatorname{logit}\big(\Pr[Y=1]\big) = \beta_0 + \beta_2 \mathbb{1}[X{=}Q_2] + \beta_3 \mathbb{1}[X{=}Q_3] + \beta_4 \mathbb{1}[X{=}Q_4] + \sum_k \gamma_k Z_k,$$

so e^{β_j} is the adjusted odds ratio for quartile j versus Q1. The adjustment set is

`age_cat, sex, race, ethnicity, education, income, alcohol, smoking, median_bmi, has_sleep_apnea, on_sleep_med, on_narcotic, has_depression, has_anxiety, has_diabetes, has_hypertension, has_pud.`

For each exposure the engine fits a *full* model and a *backward* model selected by AIC (`MASS::stepAIC`) in which the exposure is forced to remain (the selection lower scope is `~ exposure`). Each model reports the complete-case N, the ROC AUC via `pROC`, and variance inflation factors via `car`; results are rendered as merged full-vs-backward tables and stacked across exposures. All six configurations reuse this exact procedure.

21.3 Combined mutually-adjusted models

The combined file additionally fits the mutually-adjusted sleep+activity models of Section 6.3, so that, for each outcome, the sleep and activity contributions are estimated in a single equation and each is adjusted for the other domain plus the full covariate set.

21.4 Assumption diagnostics

A dedicated diagnostics routine interrogates: **(1)** independence (no duplicate `person_id`); **(2)** multicollinearity (generalized VIF > 2 or VIF > 5); **(3)** separation/sparsity (logit standard errors > 5); **(4)** influential points (Cook's distance against $4/N$); and **(5)** linearity in the logit for continuous covariates (Box–Tidwell $x \log x$ term for median BMI). It runs for both outcomes of every configuration.

22 The Six Analysis Configurations in Detail

The design crosses two outcomes with three exposure sets. The six resulting configurations are summarized in Table 16 and described individually below. Configurations sharing a column are produced by the same analysis file, because `analyze_all_outcomes()` loops both outcomes over a single modeling data frame.

ID	Outcome	Exposure set	Cohort	File
C1	Broad GERD	Sleep (8 quartiles)	$\geq$ 180 nights	Sleep and GERD
C2	Oesophagitis	Sleep (8 quartiles)	$\geq$ 180 nights	Sleep and GERD
C3	Broad GERD	Activity (7 quartiles)	$\geq$ 30 days	Activity and GERD
C4	Oesophagitis	Activity (7 quartiles)	$\geq$ 30 days	Activity and GERD
C5	Broad GERD	Combined (15 + pairs)	Intersection	Activity Sleep and GERD
C6	Oesophagitis	Combined (15 + pairs)	Intersection	Activity Sleep and GERD

Table 16: The six analysis configurations in detail.

22.1 C1 — Sleep and broad GERD

The reference configuration. On the $\geq$ 180-night sleep cohort, each of the eight sleep quartiles is regressed on `has_gerd_any` unadjusted and then adjusted for the full covariate set, with a full-vs-backward pairing per exposure. This is the direct GERD analogue of the project's IBS sleep study and the primary test of the "better sleep, lower reflux odds" hypothesis.

22.2 C2 — Sleep and oesophagitis

Identical machinery, outcome `has_esophagitis`. Because the erosive subset is rarer, case counts per quartile are smaller and confidence intervals wider; the separation diagnostic is especially relevant. Directional agreement with C1 argues that any sleep association acts on mucosal injury and not only on symptom reporting.

22.3 C3 — Activity and broad GERD

On the $\geq$ 30-valid-day activity cohort, the seven activity quartiles (steps, four activity-intensity bands, sedentary minutes, and corrected max heart rate) are modeled against `has_gerd_any`. Sedentary time and low step counts are hypothesized to associate with higher odds; the corrected max-HR exposure provides a cardiovascular-fitness proxy.

22.4 C4 — Activity and oesophagitis

The activity exposures against the erosive subset, again precision-limited by case count. Comparison with C3 mirrors the C1/C2 contrast in the activity domain.

22.5 C5 — Combined exposures and broad GERD

On the intersection cohort, all fifteen exposures are modeled separately (as in C1 and C3 but on the common sample), and the four mutually-adjusted sleep$\times$activity pairs are fit so that each domain is adjusted for the other. This isolates independent contributions: a sleep association that survives adjustment for steps, and an activity association that survives adjustment for sleep duration, are more credible than either single-domain estimate.

22.6 C6 — Combined exposures and oesophagitis

The most demanding configuration: the erosive outcome on the (smallest) intersection cohort with mutually-adjusted models. It is interpreted cautiously and primarily for consistency of direction with C5, C2, and C4.

23 The Logistic Model in Depth

23.1 Specification and estimation

Each adjusted model is a binomial generalized linear model with the canonical logit link. Writing $p_i = \Pr[Y_i = 1 \mid X_i, Z_i]$ for participant i, the model is

$$\log \frac{p_i}{1-p_i} = \beta_0 + \sum_{j=2}^{4} \beta_j \, \mathbb{1}[X_i = Q_j] + \sum_k \gamma_k Z_{ik},$$

estimated by maximum likelihood via iteratively reweighted least squares (R's `glm`). The quartile exposure enters as a three-parameter contrast against Q1, so the model makes no assumption of linearity across quartiles; a U-shaped or threshold relationship is representable.

23.2 Interpretation of the quartile contrasts

The exponentiated coefficient e^{β_j} is the odds ratio comparing quartile j to Q1, adjusted for all covariates. Reporting all three contrasts (Q_2, Q_3, Q_4 vs Q_1) reveals the *shape* of the exposure–outcome relationship rather than compressing it to a single slope. A monotone trend in $e^{\beta_2}, e^{\beta_3}, e^{\beta_4}$ suggests dose–response; a non-monotone pattern (for example, an elevated top and bottom quartile) suggests a non-linear or mixed-population effect worth further study.

23.3 Variable selection and the backward model

Alongside each full model, a backward-elimination model is selected by the Akaike Information Criterion (`MASS::stepAIC`, `direction = "backward"`) with the exposure *forced* to remain (the lower scope is `~ exposure`). The backward model is a parsimony check: if the exposure's estimate is stable between the full and backward models, the association is not an artifact of a particular covariate set. AIC balances fit against the number of parameters, discouraging overfitting in the smaller (oesophagitis, combined) cohorts.

23.4 Discrimination and multicollinearity

Each model reports the area under the receiver-operating-characteristic curve (AUC) via `pROC` as a discrimination summary, and generalized variance inflation factors (GVIF) via `car`. For factor covariates the reported quantity is $\mathrm{GVIF}^{1/(2\,df)}$, comparable across terms of differing degrees of freedom; values above 2 (or ordinary VIF above 5) flag problematic collinearity. Because the exposures within a set are correlated, they are deliberately *not* entered together in the single-exposure models; the combined mutually-adjusted models pair only one sleep and one activity exposure at a time to keep collinearity manageable.

24 Model Assumptions and Remediation

The diagnostics routine is not merely descriptive; each check is tied to a concrete remediation should it fail. Table 17 lists the assumption, the test, what a violation implies, and the recommended response.

Assumption	Test	Violation implies	Remediation
Independence	Duplicate `person_id` check	Correlated observations, understated SEs	Deduplicate joins; one row per participant.
No severe multicollinearity	$\text{GVIF}^{1/(2df)} > 2$ / $\text{VIF} > 5$	Unstable, inflated coefficients	Drop or combine collinear covariates; rely on the backward model.
No perfect separation	Logit $\text{SE} > 5$	Non-identifiable coefficients	Collapse sparse categories; consider penalized (Firth) logistic regression.
No overly influential points	Cook's distance vs $4/N$	A few rows drive the fit	Inspect flagged rows; sensitivity refit excluding them.
Linearity in the logit	Box–Tidwell $x \log x$ for BMI	Continuous covariate mis-specified	Categorize BMI or add a spline/quadratic term.

Table 17: Logistic-regression assumptions, diagnostics, and remediation.

The exposure quartiles are categorical and therefore exempt from the linearity-in-the-logit requirement; only genuinely continuous covariates (median BMI, and Charlson score if used continuously) are tested. The separation check is most consequential for the rarer oesophagitis and treated-GERD outcomes, where a covariate level may perfectly predict non-case status; the routine surfaces this as an abnormally large standard error rather than letting it pass silently.

25 Separation, Sparsity, and Model Stability

Two features of this study make naive model fitting genuinely dangerous, and together they justify an explicit safeguard rather than a post-hoc diagnostic.

25.1 The problem

First, one covariate is extraordinarily rare. `has_pud` is recorded for **137 participants** across the entire source population (Table 7) — well under one percent of the analytic cohort. Second, the erosive oesophagitis outcome is itself uncommon by construction, being a the less common of the two phenotypes in most EHR cohorts. When a rare covariate meets a rare outcome, the cross-tabulation acquires empty cells.

An empty cell is not a minor inconvenience. If no participant with peptic ulcer disease has oesophagitis, the maximum-likelihood estimate of that coefficient diverges: the fitted odds ratio runs to zero or infinity, its standard error inflates without bound, and because the coefficients are estimated jointly the instability propagates into the exposure estimate that the study actually cares about. Worse, `glm` will return such a model without error. The output looks like a result.

25.2 The guard

Before fitting, every model screens its own covariate set against its own complete-case data. A covariate is removed *from that model only* if it is constant or all-missing, if it has a single level after dropping unused ones, or if any outcome×level cell falls below `GERD_MIN_CELL` (default 1, i.e. any empty cell). Each removal is reported with its reason, and the retained and dropped sets are stored on the result object so they can be recovered later.

```
guard      <- drop_unstable_covariates(d, outcome_col, covars,
                                       context = paste0(outcome_col, " ~ ", exposure, ":"))
covars_use <- guard$keep      # constant / single-level / empty-cell terms removed
m_full     <- glm(as.formula(paste0(outcome_col, " ~ ",
                     paste(c(exposure, covars_use), collapse = " + "))),
                  data = d, family = binomial())
# ... backward AIC, AUC, VIF, Cook's distance as usual ...
```

```
list(..., covars_used = covars_use, covars_dropped = guard$dropped)
```

Listing 3: The guard, applied inside the per-exposure fitting function.

25.3 Why drop rather than penalise

Alternative treatments exist. Firth's penalised likelihood would retain the covariate and produce a finite estimate; exact logistic regression would give valid inference in very sparse tables. Both are defensible, and either could be adopted later. Dropping was chosen here for three reasons: it keeps the estimator identical to the published IBS paper, so the two studies remain directly comparable; it is transparent, in that the reader can see precisely which term was removed and why; and the removed terms are nuisance covariates rather than the exposure of interest, which is never dropped. If a future reviewer prefers penalisation, the guard is a single function and the substitution is local.

25.4 Observed behaviour

In end-to-end validation the guard behaved exactly as designed. On the broad GERD outcome, with several hundred cases, it did not fire: all covariates were retained. On the rare oesophagitis outcome it fired on every exposure model, removing `has_pud` together with several sparse demographic factors, and reported each removal. Without it those models would have returned uninterpretable odds ratios with no indication that anything was wrong.

```
[sparsity guard] has_esophagitis ~ min_asleep_quartile: dropped 5 covariate(s):
  age_cat (empty cell (min=0)); sex_birth_collapsed (empty cell (min=0));
  race_collapsed (empty cell (min=0)); ethnicity_collapsed (empty cell (min=0));
  has_pud (empty cell (min=0))
```

Listing 4: Console output from the guard during validation (abridged).

25.5 Reporting obligation

Because the guard changes the fitted model, its behaviour is a reportable feature of the analysis, not an implementation detail. If it fires on any primary model with real data, the manuscript's methods must state which covariates were removed from which models and why. The guard's output is written to the console and retained on the results object precisely so that this reporting is straightforward. A useful secondary reading is diagnostic: extensive firing on a configuration is itself evidence that the configuration is underpowered for the adjustment set, and that its estimates should be interpreted as exploratory.

26 Secondary Analysis: Restricted Cubic Splines

The primary analysis divides each exposure into quartiles. That is robust, makes no functional-form assumption, and matches the published template — but it discards within-quartile variation, imposes step changes at cut-points that carry no biological meaning, and cannot recover the *shape* of a dose–response relationship. A U-shaped association, in which both very short and very long sleep carry elevated odds, is precisely the pattern a quartile model is least able to express: Q1 and Q4 both appear elevated relative to a mid-range reference, but the model cannot say where the minimum lies or whether the rise at either end is real.

This section adds a spline analysis alongside the quartile models. It replaces nothing: both analyses are fitted on identical participants with an identical adjustment set, so any difference between them is attributable to the exposure parameterisation and to nothing else.

26.1 Methodological precedent

The procedure follows Master et al., *Nature Medicine* 2022;28:2301–2308, "Association of step counts over time with the risk of chronic disease in the All of Us Research Program" (doi:10.1038/s41591-022-02012-w). That study is the closest available precedent: the same cohort resource, Fitbit-derived exposures, EHR-derived outcomes, and Frank Harrell as senior methodologist. It also reports acid reflux among its outcomes, so the exposure–outcome pair examined here is one it examined directly.

26.2 Procedure

For each continuous exposure:

1. Fit restricted cubic splines with 3, 4 and 5 knots in separate logistic models, each adjusted for the full covariate set.
2. Select the knot count with the lowest AIC. All three candidates are fitted on the same rows, so the comparison is valid.
3. Test the exposure with Harrell's Wald "chunk test", reported as two separate quantities:
   - ***p*-overall** — jointly, are *all* spline terms zero? This asks whether the exposure is associated with the outcome at all.
   - ***p*-non-linearity** — jointly, are the *non-linear* terms zero? This asks whether that association departs from a straight line on the log-odds scale.
4. Report the effect as the odds ratio comparing the 75th to the 25th percentile of the exposure, matching the contrast Master et al. report.
5. Plot the fitted curve with a 95% confidence band across the observed range, referenced to the median.

Knots sit at Harrell's recommended quantiles (*Regression Modeling Strategies*, Table 2.3), which are the `rms::rcs()` defaults:

Knots	Quantiles
3	10, 50, 90
4	5, 35, 65, 95
5	5, 27.5, 50, 72.5, 95

26.3 The spline basis

With knots $t_1 < \cdots < t_k$, the basis is the truncated power form of *Regression Modeling Strategies* eq. 2.25: the first column is x itself and, for $j = 1, \ldots, k-2$,

$$X_{j+1} = \frac{(x-t_j)_+^3 - (x-t_{k-1})_+^3 \frac{t_k - t_j}{t_k - t_{k-1}} + (x-t_k)_+^3 \frac{t_{k-1} - t_j}{t_k - t_{k-1}}}{(t_k - t_1)^2}.$$

The linear term occupying column one is what makes the two chunk tests a clean partition: dropping columns $2, \ldots, k-1$ leaves a model linear in x, and dropping all of them removes x entirely. The basis is implemented directly rather than through `rms`, so the pipeline carries no additional dependency, and it is verified on load against `splines::ns()` — the two bases span the same function space, so a model fitted with either must produce identical fitted values. The run aborts if that check fails.

26.4 Odds ratio curves

For a model $\operatorname{logit}\Pr(Y=1)=\beta_0+\mathbf{B}(x)^\top\boldsymbol{\beta}+\mathbf{Z}^\top\boldsymbol{\gamma}$, the odds ratio at x relative to a reference x_0 is

$$\mathrm{OR}(x\,|\,x_0)=\exp\big[(\mathbf{B}(x)-\mathbf{B}(x_0))^\top\boldsymbol{\beta}\big],\qquad \widehat{\mathrm{Var}}=\mathbf{d}^\top\mathbf{V}_{\boldsymbol{\beta}}\,\mathbf{d},\quad \mathbf{d}=\mathbf{B}(x)-\mathbf{B}(x_0).$$

Every covariate is held fixed and cancels exactly, so only the spline block of the coefficient vector and its covariance sub-matrix are required. The reference and both contrast percentiles are placed on the evaluation grid, so the plotted curve passes exactly through OR $=1$ at the reference rather than merely near it.

26.5 Two deliberate departures from Master et al.

Both follow from this study's design and are stated here so that neither is mistaken for an oversight.

- **Logistic regression, not Cox.** This study is cross-sectional. There is no follow-up time and no incident-event structure, so a proportional hazards model is not defined; odds ratios replace hazard ratios. Master et al. likewise used logistic regression for their own cross-sectional screening step and reserved Cox models for the longitudinal analysis their repeated-measures design supported.
- **Complete-case, not multiple imputation.** The paper pooled results over five `aregImpute` datasets. Complete-case analysis is used here so that the spline and quartile models are fitted on exactly the same rows. Adding imputation to one arm only would confound a difference in *shape* with a difference in *missing-data handling*, which is the one comparison this section exists to make. Per-model missingness is reported so the cost is visible, and multiple imputation across both analyses remains available as a sensitivity analysis.

26.6 Stability safeguards

A spline model can converge and still be numerically meaningless. With few cases, cubic terms can yield an odds ratio of 10^5 with a confidence interval spanning twenty orders of magnitude, and — as observed during validation — such a model can carry an apparently significant p-value. Three guards prevent that reaching a table:

Guard	Action
CI ratio >1000	Estimate suppressed, row marked UNSTABLE
$\lvert\log\mathrm{OR}\rvert>\log 50$	Estimate suppressed, row marked UNSTABLE
Events per parameter <10	Estimate retained, flagged as provisional

Suppressed rows remain in the CSV with the reason recorded in `unstable_why`, so nothing is silently discarded, but they are excluded from the shape classification and displayed as `-`. The events-per-variable threshold follows Peduzzi et al. (1996); it matters more here than in the quartile analysis because a 5-knot spline spends four degrees of freedom on the exposure alone.

26.7 Validation

The implementation is verified against data with known structure before use: a simulated U-shape must be detected as non-linear with its minimum recovered in the right place; a simulated linear trend must be detected as an association but *not* as non-linear; a null exposure must trigger neither test; and the 75th-versus-25th-percentile odds ratio must agree with an independently fitted linear-logit model. All four checks, together with the basis identity against `splines::ns()`, pass.

26.8 Outputs

Written to `manuscript_output_splines/`, separate from the quartile outputs:

- `<cohort>_spline_summary.csv` — one row per outcome × exposure: selected knot count, knot locations, AIC, OR (75th vs 25th) with CI, both χ^2 statistics with degrees of freedom, both raw and multiplicity- adjusted p-values, events per parameter, and a `shape` column reading *linear*, *non-linear*, *no association* or UNSTABLE.
- `<cohort>_<outcome>_<exposure>_spline.png` — the fitted curve with its confidence band, knot locations, reference line, and the exposure distribution beneath, so that a wide interval in a sparse region is not misread as a finding.
- `<cohort>_<outcome>_<exposure>_curve.csv` — the plotted coordinates, so figures can be restyled for submission without refitting.

Multiplicity is handled exactly as in the primary analysis: Bonferroni within outcome, across exposures, governed by the same `GERD_P_ADJUST` setting.

26.9 Interpretation

The two analyses answer different questions and should be read together. The quartile models give the manuscript's headline effect estimates in the familiar form; the spline models establish whether the quartile pattern reflects a genuine monotone gradient, a threshold, or a U-shape, and where any inflection lies. Where p-non-linearity is not significant, the linear-in-log-odds summary is adequate and the quartile estimates can be reported without qualification. Where it *is* significant, the quartile table understates the relationship and the spline figure should carry the interpretation.

27 Quartile versus Spline: Did the Secondary Analysis Change Anything?

Two analyses now estimate the same associations, which raises an obvious question that should not require reading two sets of tables side by side to answer. `GERD Quartile vs Spline Comparison R9.R` answers it in one file per cohort, written at the end of the spline run.

27.1 Making the comparison fair

Three differences would make a naive side-by-side misleading. All three are eliminated by *refitting* every model inside the comparison rather than reading the two analyses' output files.

1. **The contrasts differ.** The quartile model reports Q4 versus Q1; the spline reports the 75th versus the 25th percentile. Q4 versus Q1 contrasts the middles of the top and bottom quarters, which lie further apart than P_{75} and P_{25}, so comparing the two directly would show an effect-size difference that is an artefact of the contrast rather than of the method. The comparison therefore evaluates the spline at the *median of Q1* and the *median of Q4*, so both columns answer the same question. The P_{75}-versus-P_{25} contrast is retained in its own columns.
2. **The row sets differ.** The spline analysis trims to the 1st–99th percentile; the quartile analysis does not. AIC and interval widths are not comparable across different rows, so the trim is applied to both.

3. **The covariate sets can differ.** The sparsity guard may drop different covariates in the two fits. Here it runs once and the surviving set is used for all three models.

With those controlled, any remaining difference is attributable to the exposure parameterisation, which is the only thing under test.

27.2 What each row reports

Odds ratios from both methods on the matched contrast; ΔAIC (spline minus quartile, negative favouring the spline); the ratio of confidence-interval widths on the log-odds scale, below one meaning the spline is more precise; the non-linearity p-value; and a plain-language `verdict`:

Verdict	Meaning
SPLINE ADDS: non-linear	The relationship is not log-linear. The quartile table understates it; the spline figure should carry the interpretation.
SPLINE ADDS: association only the spline detects	The quartile model missed it.
disagree	The two methods differ on significance. Investigate before reporting either.
agree; spline more precise	Same conclusion, narrower interval.
agree	Same conclusion. Report quartiles as primary; cite the spline as confirming log-linearity.
unstable	Too few cases. Suppressed on whichever side is affected; not reportable.

27.3 Why this is worth having even when the answer is "no"

A verdict of *agree* across the board is a substantive result, not a null one: it establishes that the quartile estimates are not an artefact of where the cut-points fell, and it forecloses the reviewer question "did you check the functional form?". One methods sentence then covers it.

27.4 The case this exists to catch

During validation, a simulated symmetric U-shape produced a quartile odds ratio of 1.00 (0.91–1.11) — an exactly null result — because Q1 and Q4 sit at comparable risk when the minimum lies in the middle, so the contrast between them cancels. The spline recovered the same relationship with $p_{\text{non-linearity}} < 10^{-4}$ and $\Delta\text{AIC} = -2049$. A quartile analysis alone would have reported no association where a strong one exists. This is not a hypothetical failure mode: it is the specific reason a sleep exposure warrants the check, since both short and long sleep are plausibly associated with reflux.

27.5 Stability applies to both sides

The suppression rules of the spline section are applied to the quartile estimate as well as the spline. A quartile model is no less capable of degeneracy — validation produced one with an odds ratio of 4.9×10^9 — and a table that suppressed only the spline would imply the other side was trustworthy when it was not.

28 Data Engineering, Quality Control, and Statistical Considerations

28.1 One-row-per-participant guarantee

Every derived covariate table is reduced to one row per `person_id` before joining, and the diagnostics assert no duplicate identifiers in the modeling frame. Eligibility is defined once (an `inner_join` to the filtered exposure summary), and covariates are attached by `left_join` so they never expand the row count.

28.2 Missing data

Binary EHR indicators (conditions, medications, PUD, IBS, and the outcomes) are coalesced from `NA` to `FALSE`: in EHR phenotyping the absence of a qualifying record means the participant does not meet the phenotype. Charlson score is coalesced to 0 when no components precede the exposure window. Survey/measurement covariates (education, income, alcohol, BMI) are *not* imputed; an explicit "Unknown/Missing" level is carried through and reported, so differential missingness by outcome is visible. Each adjusted model uses complete cases over its own variable set, so the effective N is reported per model and can differ across exposures.

28.3 Multiple comparisons

With fifteen exposures (each with several quartile contrasts) against two outcomes across three exposure sets, multiplicity is substantial. The analysis applies the Bonferroni correction across the models (matching the antecedent IBS paper; `GERD_P_ADJUST` selects FDR instead), and the primary multivariable emphasis is on effect sizes (aOR) and confidence intervals under a fixed, pre-specified adjustment set rather than on dichotomous significance.

28.4 Confounding structure

Each covariate blocks a plausible backdoor path: adiposity (BMI) raises intra-abdominal pressure and also shortens/fragments sleep and reduces activity; age, sex, race, ethnicity, and socio-economic status pattern both behavior and health-care coding; smoking and alcohol relax the lower esophageal sphincter and degrade sleep; depression and anxiety alter sleep and amplify symptom reporting; IBS and PUD co-travel with GERD; the Charlson index captures residual multimorbidity; and sleep-relevant medications affect measured sleep. Mediators of the exposure→GERD path (acid-suppression therapy) are excluded from the primary adjustment set to avoid over-adjustment and used only to define the treated-GERD sensitivity outcome.

28.5 Sample size and precision

No fixed sample size is imposed; precision is governed by the number of *cases*. Oesophagitis, being the erosive subset, is rarer than broad GERD, and the treated-GERD sensitivity (a further subset) is the most precision-limited. The combined intersection cohort is smaller than either single-exposure cohort. Sparse covariate×outcome cells are exactly what the separation diagnostic is designed to catch, and backward-AIC can drop unstable terms while the exposure is retained.

29 Assumed Causal Structure

The adjustment set is not chosen empirically (for example, by stepwise inclusion of "significant" covariates) but from an explicit, if informal, causal diagram. Stating the assumed structure makes the identification strategy auditable and clarifies which variables must be adjusted for and which must not.

29.1 Paths the design intends to block

The exposures (sleep, activity) and the outcomes (GERD, oesophagitis) share several common causes that open backdoor paths and would bias an unadjusted association:

- **Adiposity** → both worse sleep/less activity and more reflux (pressure gradient). Adjusted via median BMI.
- **Age, sex, race, ethnicity, socioeconomic status** → both behavior and health-care coding. Adjusted via the demographic and SES covariates.
- **Psychiatric comorbidity** (depression, anxiety) → both sleep and symptom reporting. Adjusted directly.
- **General multimorbidity and GI comorbidity** (Charlson, PUD, IBS) → both activity/sleep and reflux coding. Adjusted directly.
- **Health-care surveillance intensity** → both wearable adoption and diagnosis probability. Partially blocked by SES and comorbidity adjustment.

29.2 Paths the design intends *not* to block

Two categories are deliberately excluded from the primary adjustment set:

- **Mediators** of the exposure→outcome effect. Acid-suppression therapy plausibly lies *downstream* of reflux; conditioning on it would be over-adjustment and could induce collider bias. It is therefore used only to *define* the treated-GERD sensitivity outcome, never as a covariate.
- **Colliders.** No variable that is a common *effect* of exposure and outcome is conditioned on, since doing so would open a non-causal path.

29.3 The reverse-causation edge

The one edge the covariate set cannot address is the direct outcome→exposure arrow: nocturnal reflux fragments sleep. Adjustment cannot remove a reverse-causal effect; only design can. The post-Fitbit temporal outcome definition addresses it by requiring the outcome's EHR appearance to follow the exposure-measurement window, and its concordance with the primary estimate is reported as the key robustness check.

30 Sensitivity Analyses

1. **Distinct-date GERD** (`has_gerd_any_dates`) — the stricter "≥ 2 distinct dates" phenotype, testing robustness to same-day code inflation.
2. **Post-Fitbit temporal GERD** (`has_post_fitbit_gerd`) — restricts cases to those whose first GERD code appears ≥ 180 days *after* the first Fitbit date, so the measured exposure precedes the outcome's EHR appearance. This is the strongest available guard against reverse causation, which is a genuine concern because nocturnal reflux fragments sleep.
3. **Sleep×activity overlap** — repeats analyses on the concurrently well-measured subset.
4. **Medication-treated ("severe") GERD** (`severe_gerd_binary`) — an optional outcome flagging participants who both meet the GERD phenotype *and* have ≥ 2 PPI/H_2RA prescription dates beginning at least one day after their first GERD diagnosis, an EHR proxy for clinically managed disease. It mirrors the IBS study's "severe IBS" construction but substitutes acid-suppression pharmacotherapy.

30.1 The pharmacologics pull

The treated-GERD analysis depends on `GERD Pharmacologics Data Upload.R`, which retrieves drug exposures whose ancestor concept is a PPI (21600095) or an H_2RA (21600096) using the `concept_ancestor` double-join (the same structure as the IBS drug pull), restricted to Fitbit-sharers. The whole treated-GERD section is guarded: it runs only if `R9_gerd_drug_df.rds` exists, so the primary pipeline executes cleanly whether or not the optional pull has been performed.

31 Relationship to the IBS and Constipation Templates

The suite mirrors the project's IBS sleep study, constipation activity study, and constipation outcome upload, but global find-and-replace was insufficient; whole blocks were removed, added, or corrected.

Action	Detail
Swapped	`has_ibs` → `has_gerd_any` plus `has_esophagitis`; outcome labels; output filenames → `R9_final_analysis_*_gerd_df.rds`.
Swapped source	IBS rows from `dataset_32338507_condition_df` → the dedicated `R9_gerd_outcome` pull.
Removed	All IBS-subtype logic (subtype flags, most-frequent/tie-break assignment, mutually-exclusive `has_ibs_c/d/m/general`, multinomial models); the standalone severe-IBS block was replaced by the optional, guarded treated-GERD section.
Added	`has_esophagitis` as a second outcome; `has_pud`, `has_sleep_apnea`, `on_sleep_med` and `on_narcotic` covariates; the activity and combined analyses; the mutually-adjusted sleep+activity models; and the shared modeling engine.
Fixed	Consistent `R9_*` inputs throughout; one canonical `first_fitbit_sleep_date_df` / `first_fitbit_date_df` and `sleep_clean_df`, resolving the naming mismatches present in the IBS script.

Table 18: IBS/constipation → GERD/oesophagitis transformation.

Every GERD-specific artifact is `R9_`-prefixed or `_gerd`-suffixed, so running the GERD pipeline never overwrites an IBS or constipation artifact even though the underlying sleep-cleaning and activity-summary steps are numerically identical.

32 Reproducibility and Execution

32.1 Execution order

1. In the workspace, verify the GERD/oesophagitis seed concepts in the Cohort Builder, then run `GERD Outcome Data Upload R9.R` → `R9_gerd_outcome.rds`.
2. Ensure `GERD Analysis Helpers R9.R` is in the working directory (each analysis file `source()`s it).
3. Run any of `Sleep and GERD Analysis File R9.R`, `Activity and GERD Analysis File R9.R`, and `Activity Sleep and GERD Analysis File R9.R` — each produces both outcomes.
4. (Optional) Run `GERD Pharmacologics Data Upload.R` → `R9_gerd_drug_df.rds`, then re-run the sleep analysis to activate the treated-GERD section.

32.2 Software environment

The pipeline uses `tidyverse`, `gtsummary`, `broom`, `lubridate`, `MASS`, `car`, `pROC`, `forcats`, and `rlang`, plus `bigrquery` for the pulls. The helper installs missing packages on first load. Because all modeling lives in the helper, a change there propagates to all six configurations at once.

33 Verification and Defects Found

Because the analysis cannot be executed against real participant data outside the secure workspace, verification was designed to establish as much as possible without it. Three stages were used.

33.1 Stage 1 — schema validation

Every dataset the pipeline touches was opened and its columns and types confirmed directly, rather than inferred from the antecedent scripts. This covered the raw Fitbit, EHR, survey and person tables and all pre-built covariate objects, and produced Tables 6 and 7. Several assumptions carried over from the antecedent code proved incorrect at this stage, notably the duplicated `n_valid_days` column and the suffixed `answer_concept_id` columns.

33.2 Stage 2 — schema-exact synthetic data

Synthetic datasets were then constructed whose column names and types were checked field by field against the real schema definitions, with construction aborting on any mismatch. The synthetic data deliberately reproduced the hazards rather than an idealised structure:

- the duplicated `n_valid_days` column across two joined objects;
- the `answer_concept_id.x`/`.y` pair in the alcohol object;
- en-dashed factor levels in `age_cat` and `cci_cat`;
- character-typed `date_of_birth` and `condition_start_datetime`;
- an ultra-rare `has_pud` (20 of 3,000, mirroring the real 137) to exercise the guard;
- GERD records spanning both pre- and post-Fitbit windows, so that the two case definitions would genuinely diverge.

The outcome builder was additionally checked against four hand-constructed participants whose expected classification under each rule was known in advance: three late codes (post-Fitbit case), two early codes (case under the "ever" rule only), a single code (case under neither), and two late oesophagitis codes (case under both phenotypes).

33.3 Stage 3 — end-to-end execution

All three drivers were then run to completion against that data. The final status was **exit code 0 for all three**, producing 50 output files with zero errors and zero warnings. The mutually adjusted pair models of configuration C5/C6 were additionally exercised in isolation.

33.4 Defects found and fixed

Five defects surfaced during schema alignment and execution. Each would have either aborted the run or, more insidiously, produced plausible-looking but wrong results.
The fourth is the most instructive. It produces no error, no warning, and a complete set of output tables; the only symptom is that education and income appear entirely uninformative. The antecedent script's own comments record having encountered exactly this and worked around it by hand. Mapping from concept identifiers removes the failure mode rather than the symptom.

Defect	Consequence	Fix
`library(MASS)` masked `dplyr::select()`	Immediate failure of the first `select()` call in the pipeline; would have blocked the run entirely	MASS, car and pROC are referenced with `::` and never attached; tidyverse packages attach with dplyr last
Duplicated `n_valid_days`	Joining the wear-hours and steps summaries produced `n_valid_days.x/.y`; the descriptive table's coverage row would have failed or silently used the wrong column	The wear-hours object is slimmed to two columns before joining
Suffixed `answer_concept_id`	Contaminated the covariate bundle and collided with the education and income extracts	Only `alcohol_likert_final` is taken
Free-text socio-economic mapping	The `*_response` strings did not match the antecedent script's `case_when`; *every* participant would have collapsed to "Missing", silently destroying two covariates	Categories mapped from `answer_concept_id`, with text matching as fallback
En-dashed factor levels	`cci_cat` levels failed to match, producing 50+ level-mismatch warnings and risking silent `NA`s	Dash style normalised before factor construction

Table 19: Defects identified by schema alignment and end-to-end execution.

33.5 What this verification does and does not establish

It establishes that the code executes correctly against the real *schema*, that the join hazards are handled, that the outcome definitions classify participants as intended, that the separation guard engages when it should, and that the emitted tables and figures match the target format.

It cannot establish anything about the real data's missingness patterns, case counts, or covariate distributions, because those are not observable outside the workspace. Two console outputs should therefore be checked on the first production run. The eligible sleep cohort should print $N = 19{,}995$; a different value means a different `valid_population_sleep` object was picked up and the cohort no longer matches the published paper. And `build_gerd_outcomes()` prints case counts under both definitions; if the post-Fitbit count is very small relative to the "ever" count, that is a substantive finding about GERD ascertainment timing in this cohort and may argue for making the "ever" rule primary.

34 Anticipated Results and Interpretation Framework

Consistent with the FGID portfolio's hypothesis, we anticipate that higher sleep quantity and quality and higher physical activity will be associated with *lower* adjusted odds of both GERD and oesophagitis, and that markers of fragmented sleep or high sedentary time will be associated with *higher* odds. An aOR of, say, 0.85 for Q4 versus Q1 of `min_asleep_quartile` means that, holding the adjustment set fixed, participants in the highest sleep-duration quartile have 15% lower odds of the outcome than those in the lowest. A monotone decline across Q2→Q3→Q4 constitutes a dose–response pattern.

34.1 A worked example of reading one output row

Suppose the C1 (sleep, broad GERD) stacked table reports, for `min_asleep_quartile`, adjusted odds ratios of 0.94 (Q2), 0.88 (Q3), and 0.82 (Q4) versus Q1, with the Q4 95% CI $(0.74, 0.91)$

Exposure (high vs low)	Predicted aOR	Primary mechanism
More minutes asleep	< 1 (protective)	More clearance, less arousal, lower sensitivity.
Higher sleep efficiency	< 1	Consolidated sleep, less supine low-clearance time.
More deep / REM sleep	< 1	Restorative sleep, autonomic balance.
More minutes awake / restless	> 1 (harmful)	Fragmentation, sympathetic arousal.
More daily steps / active minutes	< 1	Weight control, vagal tone, motility.
More sedentary minutes	> 1	Adiposity, postprandial recumbency.
Higher corrected max HR	< 1 (fitness proxy)	Autonomic/cardiovascular fitness.

Table 20: Pre-specified directional predictions for the exposure–outcome associations (both outcomes). Predictions are expected to be directionally similar for GERD and oesophagitis, with acid-contact pathways potentially stronger for the erosive subset and perceptual pathways stronger for broad GERD.

and a complete-case N of, say, 18,500. This would be read as follows: participants in the highest quartile of average nightly sleep have about 18% lower adjusted odds of a GERD phenotype than those in the lowest quartile; the interval excludes 1, so the association is statistically distinguishable from the null at the conventional level; and the monotone decline $0.94 \to 0.88 \to 0.82$ across Q2–Q4 is a dose–response signature that strengthens (though cannot prove) a protective reading. The same row for C2 (oesophagitis) might show a similar direction with a wider interval, reflecting the smaller case count; a materially *stronger* Q4 effect for oesophagitis than for broad GERD would point toward an acid-contact-time pathway, whereas a weaker one would point toward a symptom-perception pathway. The backward-model column indicates whether the estimate is stable when non-contributing covariates are dropped, and the footnoted N makes the complete-case sample explicit. This single-row reading generalizes to every exposure in every configuration, because all six are produced by the same engine.

Three internal-consistency checks structure interpretation. First, *concordance across outcomes*: if an exposure association is directionally consistent for broad GERD and for the erosive oesophagitis subset, it is less likely to be an artifact of symptom-reporting behavior and more likely to reflect a mucosal-injury pathway. Second, *concordance across exposure domains*: agreement between the sleep-only, activity-only, and mutually-adjusted combined estimates indicates that neither domain is merely a proxy for the other. Third, *temporal concordance*: agreement between the cross-sectional primary estimate and the post-Fitbit temporal definition is the key guard against reverse causation. Because the design is cross-sectional, the primary estimates remain associations, not causal effects.

35 Threats to Validity

Beyond the item-level limitations below, it is useful to organize the principal threats to validity by epidemiologic type and to state how the design addresses each.

35.1 Selection bias

The cohort is restricted to participants who own a Fitbit, share it with *All of Us*, wear it enough to meet the coverage thresholds, and share EHR data. Each filter can associate with both exposure and outcome (for example, health-motivated device use). The design cannot remove this but makes it transparent through the CONSORT waterfall, and adjusts for socio-economic status, which is a principal driver of device access.

35.2 Information (measurement) bias

The exposure is objective (device-recorded) but the outcome is EHR-coded and therefore subject to detection and coding variation: participants with more health-care contact are both more likely to be coded GERD and may differ in sleep and activity. Conditioning on the Charlson index and comorbidities partially blocks this "surveillance" pathway. The ≥ 2-record and ≥ 2-distinct-date rules reduce false-positive coding at the cost of sensitivity.

35.3 Confounding

Addressed by the pre-specified adjustment set (Section 9.2), the pre-Fitbit temporal covariate rule (so covariates precede exposure), and the mutually-adjusted combined models (so each exposure domain is adjusted for the other). Residual confounding from unmeasured factors (diet, meal timing, caffeine, hiatal hernia) remains possible and is stated as a limitation.

35.4 Reverse causation

The most specific threat for reflux: nocturnal GERD fragments sleep, so a cross-sectional sleep–GERD association could run outcome→exposure. The post-Fitbit temporal outcome (`has_post_fitbit_gerd`), which requires the first GERD code to appear well after the exposure window, is the primary guard; concordance between it and the primary estimate is the key internal check.

35.5 External validity

Findings generalize to adult, EHR-sharing, Fitbit-wearing *All of Us* participants with sufficient wear time — a group that is younger, more affluent, and more health-engaged than the general population. Estimates should not be extrapolated to non-wearers without caution.

36 Limitations

- **Cross-sectional design.** Temporality is only partially addressed; primary estimates cannot establish causation, and reflux itself disrupts sleep.
- **EHR phenotyping.** Diagnosis-code outcomes miss undiagnosed or over-the-counter-managed reflux and vary with coding practice; the ≥ 2-record rule trades sensitivity for specificity.
- **Oesophagitis descendant scope.** The oesophagitis phenotype is the descendant set of concept 30753, which may include non-reflux causes (eosinophilic, infectious, pill-induced or radiation oesophagitis). Unless the seed is narrowed in the Cohort Builder, the outcome is "oesophagitis" rather than "reflux oesophagitis" specifically, and estimates should be read accordingly. The superseded approach of defining the erosive subset by concept-name match captures "GERD with esophagitis" but may miss erosive-esophagitis concepts coded outside the GERD hierarchy unless their seeds are added in the Cohort Builder.
- **Wearable selection.** Fitbit users are not representative of the general population.
- **Residual confounding.** Diet, meal timing, caffeine, body position, and hiatal hernia remain unmeasured; hiatal hernia would require a new condition pull.
- **Concept-set completeness.** GERD and drug-class concept IDs are seeds requiring Cohort Builder verification; the descendant-expansion design (Section 7.2) mitigates but does not eliminate this.

- **Covariate-timing heterogeneity.** `has_pud` is loaded from a file computed on the activity-based first-Fitbit date; the small resulting timing difference is a minor, documented caveat.

37 Future Directions and Extensions

The suite is deliberately extensible. Several extensions follow naturally from the shared-engine design, each requiring only a new modeling data frame or a new outcome column while reusing the same functions.

- **Dose–response modeling.** Replace the quartile factors with restricted cubic splines on the continuous metrics to estimate smooth exposure–response curves, using the Box–Tidwell result to decide where non-linearity matters.
- **Effect-measure modification.** Add interaction terms (for example, sleep × BMI, or sleep × sex) to test whether the reflux association differs across subgroups.
- **Erosive-severity gradient.** Extend the oesophagitis phenotype toward a graded outcome (reflux esophagitis, stricture, Barrett metaplasia) to test a monotone exposure–severity relationship.
- **Fatigue and symptom stratification.** Incorporate the fatigue-questionnaire data used in the sibling IBS work to stratify or adjust the sleep analyses.
- **Longitudinal follow-up.** Move from cross-sectional prevalence to incident GERD after the Fitbit window, strengthening the temporal argument beyond the current post-Fitbit sensitivity analysis.
- **Medication-response phenotyping.** Extend the treated-GERD flag to distinguish PPI-responsive from PPI-refractory disease, a clinically important axis of reflux heterogeneity.
- **Additional physiological exposures.** Heart-rate variability and resting heart rate, if available, would sharpen the autonomic-pathway hypothesis.

Because every extension plugs into `analyze_all_outcomes()` through the same helper, adding one does not perturb the six primary configurations, preserving the internal consistency that the architecture is designed to guarantee.

38 Integration Across the FGID Portfolio

This GERD/oesophagitis suite is the fourth member of a coordinated program that already covers irritable bowel syndrome, functional dyspepsia, and constipation, each relating the same Fitbit-derived sleep and activity exposures to a disorder of gut–brain interaction. Because the studies share eligibility rules, exposure quartilization, covariate definitions, and (now) a common modeling engine, their estimates are directly comparable. Table 21 places GERD within the portfolio.

Cross-disorder comparison is scientifically informative. If, for example, poor sleep is associated with higher odds across all four disorders, that pattern supports a shared gut–brain mechanism rather than a reflux-specific one; if the association is strongest for the erosive oesophagitis phenotype, it points to an acid-contact pathway specific to the upper GI tract. The portfolio design — identical exposures, harmonized covariates, and a common engine — is what makes such comparisons valid rather than confounded by differing analytic choices. The GERD suite's two nested outcomes add a within-disorder axis (symptom-coded vs erosive) that the other studies do not have, giving the portfolio its first built-in objective-injury comparator.

Study	Outcome(s)	Distinctive features
IBS	`has_ibs` (+ subtypes)	Multinomial subtype models; severe-IBS medication subgroup.
Functional dyspepsia	`has_fd` (strict/broad)	Rome-based phenotypes; early-satiety extension.
Constipation	`has_constipation`	Simple binary outcome; heart-rate-correction activity analysis.
GERD / oesophagitis	`has_gerd_any`; `has_esophagitis`	Two acid-related phenotypes; PUD covariate; sleep, activity, and combined exposures under one shared engine.

Table 21: The GERD/oesophagitis analysis within the functional-GI-disorder portfolio.

39 Robustness, Triangulation, and Pre-Specification

The strength of the design is less in any single estimate than in the pattern of agreement across deliberately varied analyses. This section states the pre-specified robustness and triangulation plan.

39.1 Triangulation across three axes

1. **Across outcomes.** A sleep or activity association that holds for both broad GERD and the erosive oesophagitis subset is more credible than one that appears for only the symptom-coded outcome, because the erosive phenotype reflects objective mucosal injury.
2. **Across exposure domains.** Agreement among the sleep-only (C1/C2), activity-only (C3/C4), and mutually-adjusted combined (C5/C6) estimates indicates that neither domain is merely a proxy for the other; the mutually-adjusted models are the decisive test of independent contribution.
3. **Across case definitions.** Agreement between the primary (temporal $\geq$ 2-record) definition and the stricter distinct-date, temporal, and treated-disease definitions indicates the association is not an artifact of a particular phenotype rule.

39.2 Pre-specification

The exposures, the two outcomes, the adjustment set, the reference levels, the quartile parameterization, the full-plus-backward modeling procedure, and the sensitivity analyses are all fixed in code *before* estimation, in the shared helper. Because the helper is the single source of these choices, there is no opportunity for per-configuration analytic drift or outcome-dependent covariate selection; the backward step is a parsimony check layered on top of a pre-specified full model, not a data-driven search for the reported estimate.

39.3 What would count as a robust finding

A robust protective sleep association, for instance, would show: an aOR below 1 for the top sleep quartile in both C1 and C2; a monotone quartile gradient; survival of adjustment for activity in C5/C6; concordance with the post-Fitbit temporal definition; and stability between the full and backward models. A finding meeting only some of these criteria would be reported as suggestive and interpreted with the corresponding caution. This graded standard, applied identically across all six configurations by construction, is the payoff of the shared-engine architecture.

40 Conclusion

This report specifies a complete, reproducible suite relating objectively measured sleep and physical activity to two nested EHR-defined outcomes — broad GERD and erosive oesophagitis — in *All of Us* CDR v9. Six configurations (two outcomes × three exposure sets) are executed by a single shared modeling engine that guarantees statistical identity across the sleep, activity, and combined analyses. The suite faithfully mirrors the project's IBS and constipation studies while correcting their naming inconsistencies, removing IBS-only subtype machinery, adding GERD-appropriate gastrointestinal covariates and a second erosive outcome, and extending the design to physical activity and to mutually-adjusted combined models. Together with distinct-date, temporal, overlap, and treated-disease sensitivity analyses, the design supports a robust, portfolio-consistent assessment of whether sleep and activity are associated with reflux disease and its erosive form.

A Appendix A: Output Inventory

Output .rds / object	Description
R9_gerd_outcome.rds	Raw GERD condition records (+ is_esophagitis flag).
R9_gerd_status.rds	has_gerd_any ($\geq$ 2 records) per participant.
R9_esophagitis_status.rds	has_esophagitis ($\geq$ 2 esophagitis-named records).
R9_gerd_status_dates.rds	has_gerd_any_dates ($\geq$ 2 distinct dates).
R9_gerd_post_fitbit_sleep.rds	has_post_fitbit_gerd temporal flag (sleep anchor).
R9_final_analysis_sleep_gerd_df.rds	**Sleep** modeling frame (configs C1, C2).
R9_final_analysis_activity_gerd_df.rds	**Activity** modeling frame (configs C3, C4).
R9_final_analysis_activity_sleep_gerd_df.rds	**Combined** modeling frame (configs C5, C6).
R9_comorbidity_status_sleep_gerd_df.rds	Combined comorbidity indicators.
R9_ibs_covariate_status_gerd.rds	IBS covariate indicator.
R9_cci_covariates_sleep_gerd_df.rds	Charlson score and category.
R9_bmi_covariates_sleep_gerd_df.rds	BMI covariates.
R9_medication_flags_sleep_gerd.rds	Medication-use flags.
R9_rx_post_gerd_df.rds	Treated-GERD prescription flag (optional).

B Appendix B: Concept Reference

C Appendix C: Selected Code Excerpts

```r
# Broad GERD: ≥2 GERD records (computed over ALL rows of the GERD-only pull)
R9_gerd_status <- R9_gerd_outcome %>%
  group_by(person_id) %>% summarise(has_gerd = n() ≥ 2, .groups = "drop")

# Oesophagitis (erosive) subset: ≥2 records whose concept name is esophagitis
R9_esophagitis_status <- R9_gerd_outcome %>%
  filter(grepl("esophagitis|oesophagitis", standard_concept_name, ignore.case = TRUE)) %>%
  group_by(person_id) %>% summarise(has_esophagitis = n() ≥ 2, .groups = "drop")
```

Listing 5: Two nested outcomes from one GERD pull (analysis files).

```r
source("GERD Analysis Helpers R9.R")
modeling_df <- prep_modeling_df(final_analysis_df, SLEEP_EXPOSURES)  # or ACTIVITY_EXPOSURES / union
# Runs descriptive + univariable + multivariable + diagnostics for BOTH
# has_gerd and has_esophagitis over every exposure:
results <- analyze_all_outcomes(modeling_df, exposures = SLEEP_EXPOSURES)
```

Listing 6: The shared driver: both outcomes, one exposure set.

```r
run_models_for_outcome <- function(df, outcome_col, outcome_labels, exposures, covars, ...) {
  d0 <- factor_outcome(df, outcome_col, outcome_labels)
  fit_one <- function(exposure) {
    d <- d0 %>% select(all_of(c(outcome_col, exposure, covars))) %>% drop_na()
    m_full <- glm(as.formula(paste0(outcome_col, " ~ ",
                  paste(c(exposure, covars), collapse = " + "))), d, family = binomial())
```

Purpose	Concept ID(s)	Note
GERD without oesophagitis	**4144111**	Own pull, descendant-expanded.
Oesophagitis	**30753**	Own pull, descendant-expanded.
BMI measurement	3038553	Body mass index.
IBS (covariate)	75576; 4234788; 4261072; 4057826	General + subtypes.
Peptic ulcer disease	4318534; 4134146; 4265600; 4027663	Via `R9_pud_status`.
Proton pump inhibitors	21600095	Drug-class ancestor (verify).
H_2-receptor antagonists	21600096	Drug-class ancestor (verify).
Beta-blockers	21601664	Medication class.
Calcium-channel blockers	21601765	Medication class.
Stimulants	21604753	Medication class.
Antidepressants	21604686	Medication class.
Antipsychotics	21604490	Medication class.
Anxiolytics	21604600; 21604565	Medication class.
Hypnotics	21604635; 21604653; 21604685; 21604661	Medication class.

Table 23: Consolidated concept-ID reference.

```r
    m_step <- MASS::stepAIC(m_full, direction = "backward",
                   scope = list(lower = as.formula(paste0("~ ", exposure)),
                                upper = formula(m_full)), trace = FALSE)
    # AUC (pROC), VIF (car), gtsummary full-vs-backward merge ...
  }
  purrr::map(intersect(exposures, names(d0)), fit_one)
}
```

Listing 7: Per-exposure adjusted modeling inside the engine (abridged).

```r
fit_combined_pair <- function(df, outcome_col, outcome_labels, sleep_expo, act_expo, covars) {
  d <- df; d[[outcome_col]] <- factor(d[[outcome_col]], c(FALSE, TRUE), labels = outcome_labels)
  d <- d %>% select(all_of(c(outcome_col, sleep_expo, act_expo, covars))) %>% drop_na()
  glm(as.formula(paste0(outcome_col, " ~ ", sleep_expo, " + ", act_expo, " + ",
      paste(covars, collapse = " + "))), d, family = binomial())   # each domain adjusts for the other
}
```

Listing 8: Combined mutually-adjusted model: sleep + activity together.

D Appendix D: Glossary of Key Derived Variables

Variable	Meaning
`first_fitbit_sleep_date`	Earliest valid Fitbit sleep date; anchors sleep-file covariate windows.
`first_fitbit_date`	Earliest Fitbit date (activity file: activity stream; combined file: min of both streams).
`age_at_fitbit_start` / `age_cat`	Age and its category (<30, 30–44, 45–59, 60–74, 75+) at the first Fitbit date.
`n_valid_nights` / `n_valid_days`	Valid sleep nights / valid activity days (eligibility gates).
`avg_min_asleep`, ..., `avg_sleep_efficiency`	Participant-level mean of each nightly sleep metric.

Variable	Meaning
avg_daily_steps, avg_*_active_min, avg_sedentary_min	Participant-level mean daily steps and activity-zone minutes.
avg_daily_max_hr_minute_all_days	Corrected mean of daily maximum heart rate.
min_asleep_quartile, ...	Cohort quartile (Q1 reference) of a sleep metric — a modeled exposure.
step_quartile, ..., max_hr_minute_all_days_quartile	Cohort quartile of an activity metric — a modeled exposure.
has_gerd_no_eso	Primary broad outcome: ≥ 2 GERD records with the first ≥ 180 days after the first Fitbit record.
has_esophagitis	Primary erosive outcome: ≥ 2 esophagitis-named records.
has_gerd_no_eso_dates	Sensitivity: ≥ 2 GERD records on distinct dates.
has_post_fitbit_gerd	Temporal sensitivity: first GERD ≥ 180 days after first Fitbit, then ≥ 2 codes.
severe_gerd_binary	Optional: GERD *and* $\geq$ 2 post-diagnosis PPI/H_2RA dates.
has_pud	GERD-specific gastrointestinal covariate.
has_depression ...has_sleep_apnea	Pre-Fitbit, $\geq$ 2-distinct-date comorbidity indicators.
on_beta_blocker ...on_hypnotics	Sustained pre-exposure medication-use flags.
cci_score / cci_cat	Charlson Comorbidity Index, continuous and categorical (0, 1–2, 3–4, 5+).
median_bmi / bmi_closest	Within-person median BMI; nearest BMI within $\pm$180 days.
alcohol_likert_final	Daily-drinking Likert (0 = none through 5 = $\geq$ 10/day).
smoking_binary	Ever smoked ≥ 100 cigarettes (1) vs not (0).
education_collapsed, income_collapsed	Collapsed socio-economic factors.
final_analysis_sleep_gerd_df	Sleep modeling frame (configs C1, C2).
final_analysis_activity_gerd_df	Activity modeling frame (configs C3, C4).
final_analysis_activity_sleep_gerd_df	Combined modeling frame (configs C5, C6).

E Appendix E: Results-Template Tables

The analyses are not yet run, so numeric cells are left as placeholders (“–”). These shells fix the intended reporting format; each analysis file produces exactly these tables (via `gtsummary::tbl_stack`) for both outcomes.

E.1 Adjusted odds ratios — sleep exposures (C1 broad GERD; C2 oesophagitis)

E.2 Adjusted odds ratios — activity exposures (C3 broad GERD; C4 oesophagitis)

Sleep exposure (Q4 vs Q1)	C1: broad GERD aOR	C1: broad GERD 95% CI	C2: oesophagitis aOR	C2: oesophagitis 95% CI
Minutes asleep	–	(–,–)	–	(–,–)
Minutes in bed	–	(–,–)	–	(–,–)
Minutes awake	–	(–,–)	–	(–,–)
Minutes restless	–	(–,–)	–	(–,–)
Deep sleep	–	(–,–)	–	(–,–)
Light sleep	–	(–,–)	–	(–,–)
REM sleep	–	(–,–)	–	(–,–)
Sleep efficiency	–	(–,–)	–	(–,–)

Table 25: Template: adjusted odds ratios (top vs bottom quartile) for the sleep exposures against both outcomes. Full tables also report Q2 and Q3 contrasts, backward models, N, and AUC.

Activity exposure (Q4 vs Q1)	C3: broad GERD aOR	C3: broad GERD 95% CI	C4: oesophagitis aOR	C4: oesophagitis 95% CI
Daily steps	–	(–,–)	–	(–,–)
Lightly active min	–	(–,–)	–	(–,–)
Fairly active min	–	(–,–)	–	(–,–)
Very active min	–	(–,–)	–	(–,–)
Total active min	–	(–,–)	–	(–,–)
Sedentary min	–	(–,–)	–	(–,–)
Corrected max HR	–	(–,–)	–	(–,–)

Table 26: Template: adjusted odds ratios for the activity exposures against both outcomes.

E.3 Mutually-adjusted combined pairs (C5 broad GERD; C6 oesophagitis)

Sleep term	Activity term	C5: broad GERD aOR_S	C5: broad GERD aOR_A	C6: oesophagitis aOR_S	C6: oesophagitis aOR_A
Minutes asleep (Q4)	Steps (Q4)	–	–	–	–
Efficiency (Q4)	Very active (Q4)	–	–	–	–
Deep sleep (Q4)	Total active (Q4)	–	–	–	–
REM sleep (Q4)	Sedentary (Q4)	–	–	–	–

Table 27: Template: mutually-adjusted models. aOR_S and aOR_A are the top-vs-bottom-quartile odds ratios for the sleep and activity terms, each adjusted for the other domain and the full covariate set.

E.4 Descriptive-table shell (per outcome)

F Appendix F: Per-File Walkthrough

This appendix summarizes, step by step, what each of the six deliverable scripts does.

F.1 `GERD Outcome Data Upload R9.R`

1. Pins the workspace CDR to v9 and constructs the `cb_criteria` descendant-expansion SQL for seeds 4144111 and 30753 in two separate pulls, restricted to Fitbit-sharers.
2. Exports the query to the workspace bucket, reads it back, and previews the captured GERD concepts.

Characteristic	Non-case	Case	q-value
N	–	–	
Age group, n (%)	–	–	–
Sex at birth, n (%)	–	–	–
Race / ethnicity, n (%)	–	–	–
Median BMI	–	–	–
Charlson index, n (%)	–	–	–
Depression / anxiety, n (%)	–	–	–
Peptic ulcer disease, n (%)	–	–	–
IBS, n (%)	–	–	–
Exposure quartiles, n (%)	–	–	–

Table 28: Template: descriptive table stratified by an outcome (produced for each of the six configurations), with Bonferroni-adjusted p-values.

3. Flags the oesophagitis subset by concept-name match and prints its size.
4. Saves `R9_gerd_outcome.rds` (with the `is_esophagitis` flag).

F.2 `GERD Analysis Helpers R9.R`

Defines the outcome objects, exposure lists, adjustment covariates, and label lookup, and the functions `prep_modeling_df()`, `descriptive_table()`, `univariate_table()`, `run_models_for_outcome()`, `stack_results()`, `run_diagnostics()`, and `analyze_all_outcomes()`. It installs any missing packages on load and is sourced by all three analysis files.

F.3 `Sleep and GERD Analysis File R9.R`

1. Loads R9 inputs; computes the first sleep date and age; cleans sleep records; builds participant-level sleep summaries.
2. Builds both outcomes (`has_gerd_no_eso`, `has_esophagitis`) and the sensitivity definitions.
3. Builds behavioral, comorbidity, PUD, IBS, Charlson, BMI, medication, and socio-economic covariates on the sleep anchor.
4. Applies eligibility, assembles `R9_final_analysis_sleep_gerd_df`, creates integer sleep quartiles, sources the helper, and calls `analyze_all_outcomes()` for both outcomes.
5. Runs the optional treated-GERD block if the drug file exists.

F.4 `Activity and GERD Analysis File R9.R`

Parallel structure on the activity stream: first activity date and age; steps, activity-zone, wear, and corrected max-HR summaries; the same outcomes and covariates on the activity anchor; the $\geq$ 30-valid-day eligibility; `R9_final_analysis_activity_gerd_df`; integer activity quartiles; and `analyze_all_outcomes()` for both outcomes.

F.5 `Activity Sleep and GERD Analysis File R9.R`

Builds both exposure streams, anchors covariates to the earliest Fitbit date, forms the intersection cohort, assembles `R9_final_analysis_activity_sleep_gerd_df` with both quartile sets, runs `analyze_all_outcomes()` over the union of exposures, and then fits the four mutually-adjusted sleep$\times$activity pairs for both outcomes.

F.6 `GERD Pharmacologics Data Upload.R`

Pulls PPI (ancestor 21600095) and H_2RA (ancestor 21600096) exposures via the `concept_ancestor` double-join for Fitbit-sharers and saves `R9_gerd_drug_df.rds`, enabling the treated-GERD sensitivity analysis.

G Appendix G: Reporting and Reproducibility Checklist

Item	Where addressed
Study design	Retrospective cross-sectional; Sections on design and configurations.
Setting / data source	*All of Us* Controlled Tier CDR v9; Data Sources.
Participants / eligibility	CONSORT and threshold-rationale sections.
Exposures	Sleep and activity quartile definitions; Exposure Definitions.
Outcomes	GERD and oesophagitis phenotypes; Outcome Ascertainment.
Covariates / confounders	Covariate inventory and rationale; concept-ID tables.
Missing data	Data-engineering / QC section.
Statistical methods	Logistic model, per-exposure full/backward AIC, AUC, VIF.
Multiplicity	Bonferroni correction (paper-matching; FDR selectable).
Sensitivity analyses	Distinct-date, temporal, overlap, treated-GERD.
Bias / validity threats	Threats to Validity and Limitations.
Reproducibility	Execution order; shared engine; software environment.
Data availability	Confidential; remains in the secure workspace.

Table 29: STROBE-aligned reporting and reproducibility checklist.

H Appendix H: Helper Function Reference

The shared engine (`GERD Analysis Helpers R9.R`) exposes the following objects and functions. This reference documents each one's role, inputs, and outputs.

Object / function	Role, inputs, and output
`GERD_OUTCOMES`	Named list of the two outcomes; each has `col` (the data-frame column) and `labels` (reference, case). Drives the both-outcome loop.
`SLEEP_EXPOSURES`	Character vector of the eight sleep quartile column names.
`ACTIVITY_EXPOSURES`	Character vector of the seven activity quartile column names.
`GERD_BINARY_COVARS`	Binary covariates shown as No/Yes and used as 0/1 predictors.
`GERD_LABELS`	Named character vector mapping variable names to display labels.
`GERD_ADJ_COVARS`	The adjustment covariate vector (see the Covariates section).
`prep_modeling_df(df, exposures)`	Input: a final analysis frame and its exposure names. Builds collapsed factor covariates and relevels quartiles to Q1–Q4 (Q1 reference). Output: the same frame with model-ready factors.

Object / function	Role, inputs, and output
`descriptive_table(df, outcome_col, outcome_labels, exposures, extra_cont)`	Builds a `tbl_summary` stratified by the outcome with chi-square tests and adjusted p-values.
`univariate_table(df, outcome_col, outcome_labels, exposures)`	Unadjusted ORs for exposures and covariates via `tbl_uvregression`.
`run_models_for_outcome(df, outcome_col, outcome_labels, exposures, covars, cci_label)`	Fits, per exposure, a full model and a backward-AIC model (exposure forced in); returns a named list with N, AUC, VIF, merged tables, and quick exposure rows.
`stack_results(results)`	Stacks the per-exposure merged tables into one `tbl_stack`.
`run_diagnostics(results, modeling_df, outcome_col, adj_covars)`	Runs the five assumption checks.
`analyze_all_outcomes(modeling_df, exposures, covars, cci_label, extra_cont)`	Driver: loops both outcomes, calling the four routines above; returns a nested list keyed by outcome.

I Appendix I: Data Dictionary of the Final Analysis Frames

The three `R9_final_analysis_*_gerd_df.rds` frames are one row per participant. They share the covariate and outcome columns below; they differ only in which exposure block is present (sleep, activity, or both).

Column(s)	Type	Description
`person_id`	chr	Participant identifier (unique per row).
`has_gerd_no_eso`, `has_esophagitis`	logical	The two primary outcomes.
`has_post_fitbit_gerd`, `severe_gerd_binary`	logical	Sensitivity outcomes (sleep file).
`min_*_quartile` (8)	int/factor	Sleep exposures (sleep and combined frames).
`step_quartile`, `*_active_quartile`, `sedentary_quartile`, `max_hr_*_quartile`	int/factor	Activity exposures (activity and combined frames).
`avg_min_*`, `avg_*_active_min`, `avg_daily_steps`, `avg_sedentary_min`	num	Continuous exposure means.
`n_valid_nights`, `n_valid_days`	int	Coverage counts.
`age_at_fitbit_start`, `age_cat`	num/factor	Age and category.
`race`, `ethnicity`, `sex_at_birth`	chr	Raw demographics (collapsed in the engine).

Column(s)	Type	Description
`education_response`, `income_response`	chr	Raw SES survey responses.
`alcohol_likert_final`, `smoking_binary`	num	Behavioral covariates.
`median_bmi`	num	Median BMI covariate.
`cci_score`, `cci_cat`	int/factor	Charlson index (continuous and categorical).
`has_pud`, `has_ibs`	logical	GI covariates.
`has_depression` ...`has_sleep_apnea`	logical	Comorbidity covariates.
`on_beta_blocker` ...`on_hypnotics`	logical	Medication-use covariates.

J Appendix J: Software and Package Reference

Package	Role in the pipeline
`tidyverse`	Data manipulation (`dplyr`, `tidyr`, `purrr`, `stringr`, `forcats`, `ggplot2`).
`lubridate`	Date handling for temporal covariate and BMI windows.
`rlang`	Tidy-evaluation helpers (dynamic outcome/column injection in the engine).
`gtsummary`	Descriptive (`tbl_summary`), univariable (`tbl_uvregression`), and adjusted (`tbl_regression`, `tbl_merge`, `tbl_stack`) tables with adjusted p-values.
`broom`	Tidies model output into data frames (quick exposure-row extraction).
`MASS`	Backward AIC selection (`stepAIC`).
`car`	Generalized variance inflation factors (`vif`).
`pROC`	ROC AUC (discrimination).
`bigrquery`	BigQuery execution for the outcome and drug pulls.

Table 32: R packages and their roles. The shared helper installs any that are missing on first load.

K Appendix K: Glossary of Statistical Terms

Term	Definition as used here
Odds	$p/(1-p)$ for outcome probability p.
Odds ratio (OR)	Ratio of odds between two covariate settings; 1 = no association.
Adjusted OR (aOR)	OR from a multivariable model, holding covariates fixed.
Logit	Log-odds, $\log\{p/(1-p)\}$; the link function of logistic regression.
Reference level	The category (e.g. Q1) against which other levels' ORs are computed.
Quartile	One of four equal-frequency groups (`ntile(x,4)`); Q1 lowest, Q4 highest.
Maximum likelihood	Estimation by maximizing the probability of the observed data.

Term	Definition as used here
Wald confidence interval	$\exp(\hat{\beta} \pm 1.96\,\mathrm{SE})$ on the OR scale.
Complete-case analysis	Fitting on rows with no missing model variable; the N is reported per model.
AIC	Akaike Information Criterion; balances fit and parameter count in backward selection.
Backward selection	Iterative covariate removal by AIC, with the exposure forced to remain.
AUC / c-statistic	Area under the ROC curve; probability a case outranks a non-case (0.5 = chance).
VIF / GVIF	(Generalized) variance inflation factor; multicollinearity diagnostic.
Perfect separation	A covariate perfectly predicts the outcome; flagged by very large standard errors.
Cook's distance	Influence measure; values above $4/N$ flag potentially influential rows.
Box–Tidwell test	Tests linearity of a continuous covariate in the logit via an $x \log x$ term.
FDR q-value	Benjamini–Hochberg false-discovery-rate analogue of a p-value under multiplicity.
Confounder	A common cause of exposure and outcome; adjusted for.
Mediator	A variable on the causal path from exposure to outcome; not adjusted for (over-adjustment).
Collider	A common effect of two variables; conditioning on it induces spurious association.
Reverse causation	Outcome influences the measured exposure; addressed by the temporal outcome.

End of report. This document describes code and methodology only; it contains no individual-level *All of Us* data.

L Appendix L: Design History and Protocol Amendments

The body of this report describes the study as it now stands. This appendix records how the specification changed during development, so that a reader comparing against an earlier draft, or against output files produced before a change, can see what moved and why. Nothing here is current methodology; it is provenance.

L.1 Outcome definition, amended twice

1. **Original.** `has_gerd_no_eso` seeded on concept **4144111** ("GERD without oesophagitis"), pulled in parallel with oesophagitis (**30753**). Abandoned once the Cohort Builder showed substantial overlap between participants carrying 4144111 and participants carrying oesophagitis codes: participants accumulate both kinds of code over time, so the concept's label does not identify a non-oesophagitic group.
2. **Intermediate.** Broad GERD (**318800**) with a "GERD without oesophagitis" group derived by excluding oesophagitis carriers, giving three modelled outcomes and nine configurations.

3. **Current.** Broad GERD and oesophagitis modelled directly, as two outcomes over three exposure sets. The derived non-oesophagitic flag is retained descriptively but is not modelled, does not appear in any table, and consumes no multiplicity budget.

Output files carrying a `gerd_no_eso` prefix are from a run under design 2 and should be deleted before re-running.

L.2 Data source, amended

The study originally assembled its analysis frames from the project's existing `.rds` objects, which are the same objects behind the published IBS paper. It now builds everything from the three Workbench datasets described in the Data Sources section. The two studies therefore share no denominator, and a GERD number cannot inherit an IBS-era eligibility decision.

L.3 Outcome pull, amended

The original pull queried the *All of Us* production CDR project directly and exported through Cloud Storage. In the migrated workspace that project lies outside the VPC Service Controls perimeter and every request failed. The replacement queries the BigQuery dataset attached to the workspace as a Resource, which is inside the perimeter, and downloads directly.

L.4 Confounding control, amended on expert review

Sleep medication, obstructive sleep apnoea and opioid analgesics were added to the adjustment set; foregut surgery and upper gastrointestinal cancer became exclusions rather than covariates; `has_ibs` was dropped as constant in this concept set; and the partial comorbidity index was removed from the default adjustment set because it double-counts diabetes and peptic ulcer disease. Each change and its rationale is documented in the Covariates and Cohort Construction sections.

L.5 Analysis, extended

The restricted cubic spline analysis and the quartile-versus-spline comparison were added after the primary quartile analysis was complete. Both are additive: no existing analysis file was modified, and the quartile results are unchanged by their presence.